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Breaking the Cycle of Violence: A Self-Attachment Technique Game for Transforming Conflicts

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Abstract

The cycle of violence is a widely observed phenomenon where individuals who experience violence, particularly in childhood, are more likely to perpetuate violence later in life. This vicious cycle underscores the urgent need for therapeutic interventions that can break this destructive pattern. Psychotherapy has long been recognized as a critical tool for addressing trauma and emotional dysregulation, yet traditional therapy remains inaccessible for many due to high costs and limited availability.

In response to these challenges, the Self-Attachment Technique (SAT) was developed as a self-administered psychotherapy aimed at re-parenting the “childhood self”, promoting emotional healing, and fostering resilience. SAT offers an innovative, accessible approach to managing trauma and breaking the cycle of violence by enabling individuals to form secure internal attachments.

This project integrates SAT with a 2D narrative-driven game that focuses on the cycle of violence arising from genocides. The game serves a dual purpose: it functions as a therapeutic tool for individuals who have experienced genocide-related trauma, offering SAT exercises that encourage emotional self-regulation and conflict resolution. By engaging with these practices, players are guided towards breaking free from their trauma and the cycle it creates.

On the other hand, the game tries to function as an educational tool, by providing an immersive narrative experience in which players navigate the aftermath of violence and genocide. Through the artificial trauma, the players, even without the necessity of having experienced a real one, are hoped to understand the trauma better and reflect on themselves about what role they can play in resolving the conflict. Thus forming a more peaceful global society.

By combining narrative-driven gameplay with SAT exercises, this project demonstrates the potential of video games to serve as both a means for emotional healing and an educational tool to address the broader social issue of violence perpetuation.

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Contents

1	Introduction	1
1.1	Background & Multivation	1
1.2	Contributions	2
1.3	Privacy & Ethics	3
1.4	Legal Issues	4
1.5	Report Outline	4
2	Literature Review	6
2.1	Attachment Theory	6
2.2	Attachment Objects	7
2.3	Self-Attachment Technique (SAT)	8
2.4	Narrative-Driven Gameplay	9
2.4.1	Definition & Advantage	9
2.4.2	Techniques	10
2.4.3	Player Evaluation	10
2.4.4	Conclusion	11
2.5	Image-to-Image	11
2.5.1	Stable Diffusion	11
2.5.2	ComfyUI	12
2.5.3	Low-Rank Adaptation (LoRA)	13
2.5.4	ControlNet	14
2.6	Serverless Computing	15
2.6.1	Mechanism of Serverless Computing	15
2.6.2	Advantages	15
2.6.3	Disadvantage(s)	15
2.6.4	Conclusion	16
2.7	Unity	16
3	Case Study: <i>The Last of Us Part II</i>	18
3.1	Narrative and Thematic Ambitions	18
3.2	Criticisms of Moral Lessons	19
3.3	Forgiveness vs. Revenge	19
3.4	Healing as the Key to Breaking the Cycle	20
3.5	Conclusion	20

4	Story Design	22
4.1	Storyline Overview	22
4.2	Character Development	23
4.2.1	Ian	23
4.2.2	The Protagonist	24
4.3	Underlying Themes & Designs	25
4.3.1	Genocide	25
4.3.2	The Cycle of Violence	26
4.3.3	Revenge vs. Forgiveness	26
4.3.4	Healing & Empathy	27
4.3.5	Justice & Morality	27
4.4	Interactive Storytelling	27
4.4.1	Interaction with NPCs	27
4.4.2	Scripted Events and Cutscenes	28
4.4.3	Environmental Storytelling	28
4.4.4	Choice-based events	29
4.5	Summary	31
5	SAT Integration	32
5.1	SAT protocols used	32
5.2	“Gamified” SAT Exercises	33
6	Facial Expression Changer	35
6.1	Workflow Design	36
6.1.1	Basic Image-to-Image workflow	36
6.1.2	Final Workflow	38
6.1.3	Other Workflows	40
6.2	Front-end design	41
6.3	Cloud Server Setting	42
6.3.1	Evaluation	44
7	Game Design & Software Engineering	45
7.1	Game Design	45
7.1.1	General Style Choice	45
7.1.2	Character Creator	46
7.1.3	Tips Button/Panel	48
7.1.4	Photo Button/Panel	48
7.1.5	Cursor Implication	48
7.2	Software Engineering	49
7.2.1	Scene Loading Mechanism	49
7.2.2	Centralized Event Handling System	51
7.2.3	Animation Sequence and Manager	53

8	User Evaluation	56
8.1	Evaluation Questionnaire	56
8.1.1	Part I: Pre-Game Evaluation	57
8.1.2	Part II: Post-Game Evaluation	58
8.2	Conclusion	60
9	Conclusion & Future Work	61
9.1	Conclusion	61
9.2	Future Work	62
	Bibliography	63

Chapter 1

Introduction

1.1 Background & Multivation

The universal cycle of violence is a widespread phenomenon observed across various contexts and cultures. “Old wisdom” has long revealed the retaliatory nature of human behaviour, encapsulated in the phrase “tit for tat.” This cycle suggests that people who experience violence are more likely to perpetuate it. In her research, C.S. Widom demonstrated that children who suffered from abuse or neglect are significantly more likely to be arrested for delinquency, adult criminality, and violent criminal behaviour when they grow up [1]. Similarly, groups who experienced massive organized violence, such as genocide, can exhibit increased potential for violence under certain conditions, as a result of their diminished capacity to trust others and their belief that violence might be necessary for self-protection [2]. The idea is further supported by de Zulueta. In her book, she argues that trauma experienced in childhood can lead to violent behaviour in adulthood [3].

Despite the troubling evidence of this cycle, it is crucial to recognize that such a grim fate is not inevitable. Widom notes that while victims tend to have a higher probability of transmitting violence, this relationship can be altered through early interventions that assist victims in overcoming their trauma [1]. Felicity also stressed the importance of therapeutic interventions in addressing the trauma by referring to them as a means to break the cycle of violence [3].

However, traditional mental therapy can be impractical for many. The high cost of psychotherapy poses a significant barrier for people who are not wealthy enough [4]. Additionally, developing countries often have less focus on mental health issues, resulting in fewer trained therapists available. According to a study, therapist density (ratio of therapists to the population) in developing countries is only about 0.5% of that in developed countries [5]. Ironically, research has also shown that lower socioeconomic groups are more likely to experience mental disorders [6], which is even more critical for those who have endured extreme trauma, such as genocide.

Given these barriers, there is a pressing need for alternative and more accessible

forms of therapy that can reach those who need it most. The Self-Attachment Technique (SAT), originally introduced in [7, 8], offers a promising solution. As a self-administered psychotherapy technique based on Attachment Theory, SAT has been shown to effectively promote self-awareness and emotional management by having individuals “re-parent” their “childhood self” – a representation of their emotionally damaged self, while their “adult self” provides care and guidance [9]. This process aims to create new neural pathways that mimic the effects of secure attachment, fostering personal healing and resilience. According to Attachment Theory, secure attachment in childhood typically leads to a stable and positive self-image and the ability to regulate emotions effectively, whereas insecure attachment can result in difficulties with emotion regulation and an increased vulnerability to mental health disorders later in life [10, 11].

An experiment was conducted with thirty Iranian women who had a history of clinical depression and/or anxiety to assess the effectiveness of this approach. After participating in eight SAT sessions over twelve weeks, both their anxiety and depression levels decreased significantly [12]. The latest research also shows the intervention of SAT can enhance one’s well-being [13]. The experiment results demonstrate SAT’s effectiveness in emotion regulation and suggest great potential for further applications.

However, beyond its therapeutic use, SAT also holds significant potential as an educational tool aimed at preventing the cycle of violence. Teaching individuals – both those with and without direct experience of trauma – skills such as empathy, emotional regulation, and non-violent conflict resolution can effectively prevent the very start of violence. By integrating SAT principles into education, individuals can learn to manage conflicts peacefully, reducing the likelihood of engaging in violent behaviour.

SAT’s therapeutic foundation makes it ideal for broader educational applications. Techniques designed for emotional healing, such as re-parenting the childhood self and fostering self-compassion, can be used to teach individuals how to recognize and manage their emotions. This can help not only in healing trauma but also in preventing violent behaviour by promoting emotional intelligence and conflict resolution.

1.2 Contributions

By integrating the SAT with a narrative-driven game, this project aims to first serve as a therapeutic tool for the people traumatized by genocide and help them regain mental well-being, keeping them from becoming part of the violent cycle.

This project also comes up with an innovative approach to extending the benefits of SAT beyond its traditional applications, aiming to create a broader impact. Inspired by the concept of creating a virtual avatar for therapeutic interactions, this project

seeks to create a “virtual childhood trauma” experience through immersive storytelling and narrative-driven gameplay. Much like how readers or gamers can feel a deep sense of immersion in novels or games, this project seeks to evoke empathy in players by immersing them in the experience of extreme childhood trauma – in this case, genocide. By developing a narrative-driven game centred around genocide and incorporating SAT exercises, the project enables players to confront and process emotional challenges in a controlled environment. Even without having experienced such trauma in their own lives, players are encouraged to understand and empathize with the effects of trauma. With SAT integrated into the gameplay, the game fosters empathy, promotes emotional regulation, and aims to cultivate a more compassionate global society.

In this project, a demo game is developed as a trial of this idea. The game follows the story of two genocide survivors, one of whom being the player-controlled protagonist, and explores how their different choices lead to divergent paths. By building an immersive game world with carefully designed gameplay and narrative, the project seeks to leverage player empathy to simulate the trauma as if it were happening to them. The SAT exercises, which are introduced after the in-game genocide event, will help players develop self-awareness and emotional management skills, while also deepening their understanding of the psychological roots of conflict. The story will then fast-forward to several years later, allowing the players to see what the trauma has made the two characters. Then, as a test of the mechanism’s effectiveness, the game reaches its climax in a critical choice: while the other victim is consumed by revenge and hatred, perpetuating the cycle of violence, the player faces the decision to either join this cycle or stand against it to break it. User evaluation shows this game has the potential to serve as both an educational and therapeutic prototype, demonstrating the potential of this approach to promote healing and personal growth.

Additionally, a stable diffusion workflow is implemented to generate dynamic images with various emotional expressions based on user-uploaded childhood photos. This tool, hosted on a GPU cloud server, serves as a substitute for VR avatars in the 2D game setting and enhances the SAT exercises by providing personalized, visual feedback in real-time. This innovation adds an extra layer of immersion and strengthens the connection between the player and their in-game experience, helping to make the SAT exercises more impactful.

1.3 Privacy & Ethics

This project, including the developed game, is designed to run entirely on the player’s local machine, ensuring that no data will be collected from the players. Photos uploaded by the players will only be used for generating images for future SAT exercises within the game. The uploaded photos will not be stored or persisted on the server; they will be automatically deleted as part of the self-destruction process of the cloud function instance once the task is complete. This approach ensures that players’ pri-

vacy is fully protected, and their data is not misused.

From an ethical perspective, while the game contains scenes involving violence and some bloody elements, its primary function is to serve as both an educational and therapeutic tool. The depiction of violence is intended to help players understand the negative impact of such behaviour, the trauma associated with genocide, and the reasons behind the emergence of the cycle of violence. Although the game includes detailed depictions of violent behaviour, it does so with the clear intention of promoting non-violence and empathy. The project in no way advocates or glorifies violence; instead, it seeks to foster a deeper understanding of its harmful effects and to encourage players to break the cycle of violence.

1.4 Legal Issues

Most of the art assets used in this game are not originally created. Instead, they were obtained either through purchased assets or open-source resources. Typically, the assets used in *Stardew Valley* were obtained from https://www.sprisers-resource.com/pc_computer/stardewvalley/, which falls under the Apache License. The assets used for bloody scenes were purchased from <https://admurin.itch.io/npc-bundle-7>.

As this project is non-commercial, these assets are used in accordance with their respective licenses, strictly for development purposes.

1.5 Report Outline

In this report, Chapter 2 provides a comprehensive literature review of the key concepts that form the foundation of this project, including the Self-Attachment Technique (SAT), narrative-driven gameplay, and Stable Diffusion.

Chapter 3 focuses on the analysis of existing narrative-driven games, with a particular emphasis on *The Last of Us Part II* [14], a game that explores the theme of breaking the cycle of violence, typically by choosing forgiveness over revenge. Despite winning the Game of the Year in 2020, the game remains highly controversial among players, with many resisting the message it sought to convey. This game will be studied as a negative example and lessons will be drawn to ensure that the idea of breaking the cycle of violence is presented in a more engaging and acceptable way for the public in this project.

Chapter 4 provides a detailed illustration of the story design, incorporating lessons learnt to ensure that the narrative is as realistic and immersive as possible.

Chapter 5 introduces how SAT works in general and how it has been adapted to better fit the theme and purpose of the game.

In Chapter 6, the design and integration of a Stable Diffusion-powered facial expression model are discussed. This chapter covers the workflow design, performance analysis, cloud server hosting, and its integration within the project.

Chapter 7 focuses on the software design of the game itself. The first part covers design elements aimed at enhancing user experience, while the second half focuses more on the software engineering techniques used, highlighting mechanisms that make the project different from standard Unity development practices.

Chapter 8 presents the results of an early user evaluation, providing insight into player feedback and how it has informed project improvements.

Finally, Chapter 9 concludes the project by summarizing its results and providing potential future work.

Chapter 2

Literature Review

2.1 Attachment Theory

Attachment theory, initially developed by John Bowlby[15] and further expanded by Mary Ainsworth, explains how early relationships with primary caregivers shape an individual's emotional and social development. According to the theory, human beings have an instinct to mimic their primary caregivers at an early age in order to feel secure and safe. Thus, different attitudes of primary caregivers can result in various outcomes. Currently, the quality and dynamic of the relationship between the caregiver and the child are divided into four categories of attachment: secure attachment and three types of insecure attachments (avoidant, anxious/ambivalent, and disorganised)[16]:

- **Secure Attachment:** Characterized by distress when the caregiver leaves and comfort upon their return. Securely attached children feel confident exploring their environment.
- **Avoidant Attachment:** Children show little distress when the caregiver leaves and avoid them upon return. This style often develops from caregivers who are emotionally unavailable or unresponsive.
- **Anxious/Ambivalent Attachment:** Marked by intense distress when the caregiver leaves and ambivalence upon their return. These children may be clingy and dependent, resulting from inconsistent caregiving.
- **Disorganized Attachment:** This style is seen in children who exhibit a lack of clear attachment behaviour. It often results from caregivers who are frightening or traumatizing.

Secure attachment in childhood, achieved through consistent and responsive caregiving, leads to healthy emotional regulation and positive interpersonal relationships in adulthood. Conversely, insecure attachment, resulting from neglectful or inconsistent caregiving, can result in difficulties in emotional regulation, increased anxiety, and a higher risk of mental health issues.

Attachment theory is not limited to the childhood phase; it continues to shape an individual's relational and emotional experiences throughout adulthood. Adults with a secure attachment typically feel comfortable with intimacy and autonomy within their relationships. Instead of caregivers, such as parents, the attachment figures in adulthood are often romantic partners, friends, and others in society.

In general, attachment theory provides a comprehensive framework for understanding the development and impact of early relationships on a person's emotional and social life. It emphasizes the importance of responsive caregiving in fostering secure attachment, which is foundational for mental health and relational well-being throughout life.

2.2 Attachment Objects

Attachment objects are items that provide emotional comfort and security when secure attachments are not available. Such attachment substitutions are particularly useful during times of extreme stress or anxiety, such as the loss of loved ones, significant shocks, or post-traumatic stress disorder (PTSD) [17]. These objects often hold significant sentimental value and serve as tangible representations of security and attachment for individuals.

The concept of attachment objects has been explored extensively in various contexts, demonstrating their role in emotional regulation across different stages of life. In ethology, research by Harry Harlow on rhesus monkeys showed that infant monkeys preferred cloth-covered surrogate mothers over bare wire ones, emphasizing the importance of tactile comfort and emotional support [18]. This preference for a comforting object, even when it does not provide nourishment, highlights the innate need for emotional security.

In human development, Donald Winnicott introduced the idea of transitional objects, such as blankets or soft toys, which children use to self-soothe and feel secure. These objects help children transition from dependence on their caregivers to greater independence by internalizing the comfort provided by these objects [19]. The attachment to these objects often persists into adulthood, where they continue to serve as sources of comfort during times of stress.

Furthermore, the use of attachment objects extends into spiritual and religious practices. Many individuals use religious figures or deities as attachment objects to regulate their emotions and feel secure. Studies have shown that religious practices, such as prayer, can activate the same neural pathways as those involved in attachment bonds with caregivers or romantic partners, providing a sense of stability and peace during difficult times [17].

In adulthood, attachment objects can take various forms, such as photographs, mementoes, or even the presence of loved ones. These objects help bridge the gap when

direct support from significant others is not available, ensuring a continuous sense of security and emotional regulation. For example, a widow might find comfort in a piece of jewellery that belonged to her late spouse, or a soldier might carry a family photograph during deployment.

The ability of attachment objects to provide emotional stability is rooted in their symbolic representation of security and attachment. They offer a tangible means of accessing the emotional comfort that would typically be provided by a secure attachment figure. This function is crucial in helping individuals manage stress and anxiety, contributing to their overall emotional resilience and mental health.

2.3 Self-Attachment Technique (SAT)

The Self-Attachment Technique (SAT) is a holistic therapeutic approach designed to enhance emotional regulation, self-awareness, and overall psychological well-being, developed by the Algorithmic Human Development group at Imperial College London[9]. It integrates principles of attachment theory, neuroplasticity, and cognitive-behavioural therapy to help individuals with insecure attachment histories form secure internal attachments.

SAT involves a series of self-administered protocols aimed at creating an internal nurturing relationship between an individual's "adult self" and their "childhood self" [8]. As mentioned in the previous sections, early attachment experiences significantly impact an individual's capacity for emotional regulation and mental health. By replicating the dynamics of a secure child-caregiver relationship within oneself, SAT seeks to rewire the brain's neural pathways through neuroplasticity, thereby fostering resilience and emotional stability.

The term "self-" in the Self-Attachment Technique underscores the self-administered nature of the therapy. Unlike traditional therapeutic methods that rely on external relationships (such as with a therapist or caregiver), SAT empowers individuals to take an active role in their own healing process. This self-directed approach involves the adult self taking on a nurturing, supportive role for the child, mirroring the care and empathy that would typically be provided by a secure attachment figure [8]. What is more, internalizing the role of the caregiver ensures that the supportive figure is always available. This can be particularly beneficial for individuals who have experienced unreliable or inconsistent caregiving in the past, as it guarantees a stable and continuous source of emotional support.

Typically, the SAT protocol involves four stages [8]:

- **Introduction to Secure Self-Attachment Therapy:** In this initial stage, individuals are introduced to the scientific foundations and underlying principles of SAT, including the Attachment Theory, regulation theory, and their neural correlates etc. It is expected that the volunteer needs to be dedicated and

motivated for the method to be fully effective.

- **Connecting with the Child:** This stage focuses on establishing a relationship with the childhood self, which represents the emotionally vulnerable part of oneself that holds past traumas and unmet needs. The participants are expected to visualize their inner child as concretely as possible by looking at their childhood photos, past memories and through a set of exercises so that it is easier for them to develop empathy and compassion.
- **Making an Affectional Bond with the Child:** In this crucial stage, individuals work on creating a deep and affectionate bond with their childhood self. Based on the hypothesis of narcissism within all individuals, the participant is supposed to love the child and act as a secure attachment for him/her. Singing is preferred to barely speaking in the imaginative bond-making, as it is proved that singing leads to more dopamine release, and is more likely to motivate the participant during the process.
- **Developmental Retraining and Re-Parenting the Child:** The final stage focuses on reparenting the child through structured interactions that emulate those of a “good enough” parent. The goals are to minimize negative emotions and maximize positive ones. By revisiting and reprocessing traumatic childhood experiences, and having the adult self comforting the child, such as cuddling, the participant is hoped to have a more positive emotion related to those sad events.

The SAT offers a unique and effective approach to emotional healing by leveraging the power of self-compassion and internal attachment. Through fostering a nurturing relationship with one’s childhood self, SAT aims to rewire traumatized neural pathways and promote emotional resilience, providing a robust framework for overcoming past attachment insecurities and enhancing overall psychological well-being.

2.4 Narrative-Driven Gameplay

2.4.1 Definition & Advantage

Narrative-driven gameplay focuses on storytelling as a core component of the gaming experience, emphasizing character development, plot progression, and player choice to create immersive and emotionally engaging narratives [20]. Unlike action-oriented or mechanics-focused games, narrative-driven games prioritize the story and character arcs, often integrating interactive elements that allow players to influence the game’s direction based on their decisions. This approach has proven effective in fostering empathy, encouraging moral reasoning, and facilitating educational or therapeutic outcomes, making it a valuable tool in game-based learning and mental health interventions.

2.4.2 Techniques

To make the narratives engaging, such games often come with the following features:

Character Development

Character development is a critical aspect of narrative-driven gameplay, where characters with rich backstories, distinct personalities, and complex motivations enhance the storytelling experience. Research has shown that well-developed characters help players form emotional connections, making the narrative more engaging and relatable [21]. For instance, in “The Last of Us” [22] the complex relationship between Joel and Ellie adds depth to the story, fostering empathy from the player.

Interactive Storytelling

Interactive storytelling, where player agency shapes the narrative, leading to multiple possible endings and personalized experiences is another important feature that makes narrative-driven games engaging and immersive. Research indicates that such interactivity increases player engagement and emotional involvement, as players feel more connected to the story when their decisions matter [23]. Games like “Life is Strange” [24] use branching narratives to explore themes of morality, consequence, and human relationships, demonstrating how player choices can drive the story in meaningful ways. Typical interactive storytelling includes branching narratives, environmental storytelling, choice-based storytelling, scripted cutscenes, etc.

Immersive World-Building

Creating a rich, detailed game world with its history, culture, and conflicts enhances immersion, drawing players deeper into the narrative. World-building adds depth to the story, making the game environment feel alive and real, which can significantly increase player immersion and emotional connection [20]. In “The Witcher 3: Wild Hunt” [25], the game’s interesting fantasy world and intricate lore allow players to explore and engage with the narrative on multiple levels, providing a more holistic and immersive storytelling experience.

2.4.3 Player Evaluation

Research has shown that narrative-driven games can significantly enhance player experience by fostering a sense of immersion and emotional connection to the game world and its characters [26]. Players often report feeling more engaged and invested in narrative-driven games because the story gives context and purpose to their actions within the game [27]. This type of engagement is often referred to as “narrative immersion,” where players lose themselves in the story and identify with the characters they are controlling.

2.4.4 Conclusion

Incorporating narrative-driven gameplay into this project is essential for creating a deeply immersive and emotionally resonant experience, which is crucial for simulating a “virtual trauma” that players can engage with on a personal level. This immersion is vital for setting the stage for the subsequent Self-Attachment Technique (SAT) process, as it allows players to explore complex themes such as trauma, empathy, and the cycle of violence in a meaningful way. By leveraging storytelling elements, the game fosters emotional connections and promotes ethical reflection, helping players internalize these themes. The literature reviewed in this section demonstrates that narrative-driven games are particularly effective in achieving these outcomes, supporting the project’s educational and therapeutic goals.

2.5 Image-to-Image

Image-to-Image is the task of transferring an input source image to match the style or other characteristics of a target image [28]. In this project, this task is specifically desired for transferring the player’s input image to a one with different emotion, providing a naive substitute to VR for self-avatar generating, which can provide a great enhancement for the latter SAT exercises. Stable Diffusion [29] is specifically selected for this task, thanks to its open-source nature and lively community.

2.5.1 Stable Diffusion

Stable Diffusion is a popular deep learning model for image generation, first released in 2022 by StabilityAI [29]. Unlike earlier generative models like GANs (Generative Adversarial Networks), Stable Diffusion models utilize a diffusion-based approach that progressively refines a noise-corrupted image towards a data distribution that aligns with a given text prompt. Besides generating images conditioning on the given text prompts, it also performs well in other tasks guided by text input, including image-to-image translation [30].

Stable Diffusion operates by iteratively applying noise to an image and then reversing this process to generate new data samples. As shown in Figure 2.1 from its original paper [29], the model then learns to predict the noise added at each step, gradually refining a noisy image into a coherent one. This denoising process is conditioned on the input data—in the case of Stable Diffusion, textual descriptions—which guides the model to generate images that align with the specified text. The process typically involves two stages: the forward diffusion process, where noise is incrementally added to a data sample, and the reverse diffusion process, where the model learns to remove this noise step-by-step to reconstruct the original data [29].

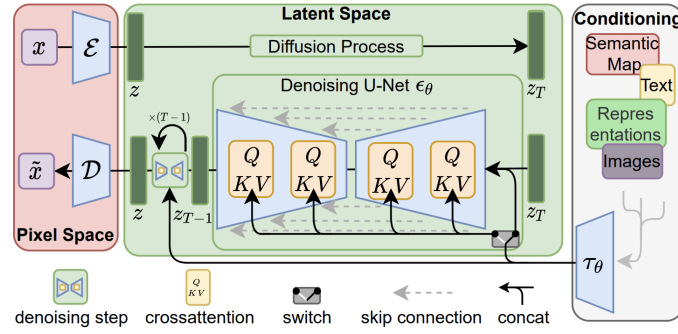


Figure 2.1: General structure of Stable Diffusion Model from [29]

Advantages

The diffusion-based approach used by Stable Diffusion offers several advantages over other generative models:

- **Open-Source Accessibility:** Its open-source nature makes it much easier to leverage the pre-trained model, allowing the developers to use it without going through the resource-intensive or time-consuming training process. This also makes it more convenient for the developers to fine-tune the model for specific tasks.
- **Community Contributions:** The lively community of Stable Diffusion has made numerous contributions to the technology. This collective effort can lead to a more robust and versatile tool that benefits all users.
- **Latent Space Control:** Stable Diffusion's latent space is more straightforward to control because it can be directly influenced using text prompts. Compared to other models with image-to-image capabilities, such as StyleGANs, which require special mechanisms to explore the latent space and manually search for editable features, Stable Diffusion's text-based control makes the image-to-image process more user-friendly and accessible.

2.5.2 ComfyUI

In this project, a modular diffusion model GUI and backend – ComfyUI [31] is used for exploring the Stable Diffusion workflow.

ComfyUI [31] is an open-source user interface designed to simplify the process of working with deep learning models, particularly those used for image generation tasks like Stable Diffusion. This platform provides an accessible, modular, and visually intuitive environment that enables users, both novice and experienced, to manipulate, customize, and deploy AI models with ease.

The interface makes complex model workflows more user-friendly and visually intuitive by offering a graphical user interface that allows users to drag and drop various

components, such as neural network layers, image processing steps, and other operations, to construct and fine-tune machine learning models. This modular approach contrasts sharply with traditional, code-heavy methods, making ComfyUI particularly appealing to users who prefer visual programming or those who may not have extensive coding experience in large models.

ComfyUI typically provides several appealing advantages over the traditional coding workflow:

- **Accessibility:** ComfyUI lowers the barrier of the image generation tasks with Stable Diffusion by providing a more intuitive environment for building and experimenting with models.
- **Ease of Integration:** ComfyUI supports a wide range of models, including Stable Diffusion, Computer Vision models like YOLO, and even Large Language Models (LLMs). This versatility allows users to experiment with different models and techniques within a single platform. What is more, the platform supports a wide range of plugins and extensions, enabling users to add new functionalities or integrate external tools as needed.
- **Efficiency:** By simplifying the model-building process, ComfyUI allows users to focus on experimentation and refinement rather than being buried by coding and debugging. This efficiency can accelerate research and development, leading to faster iterations and progress. Since this project is not particularly aimed at Stable Diffusion, this efficiency is much appreciated.

The ComfyUI is also a WebUI application in itself, making it easier for the later deploying process and API calls. Although its versatility has made the whole application considerably large in size, it is still an ideal tool for doing workflow exploration and as a backend server.

2.5.3 Low-Rank Adaptation (LoRA)

Low-Rank Adaptation (LoRA) is a technique designed to efficiently fine-tune large pre-trained models by significantly reducing the number of trainable parameters [32]. Originally focusing on Large Language Models (LLMs), LoRA is proven to be working well with other large pre-trained models, including Stable Diffusion. By introducing low-rank matrices into the layers of neural networks, LoRA enables the efficient adaptation of models without requiring the full retraining of all parameters, making it both time and resource-efficient [32].

As shown in Figure 2.2, which is from [32], LoRA works by freezing the original weights of a pre-trained model and learning additional small, low-rank matrices that are added to the model's layers. Compared to the full fine-tuning, where $d \times d$ parameters need to be stored and re-calculated, the LoRA allows to do the same thing by only changing $2 \times r \times d$ parameters, with $r \ll d$, which drastically speeds

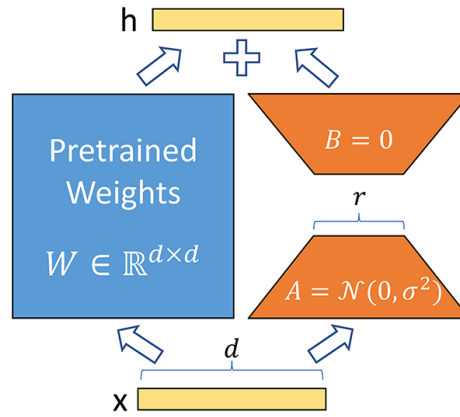


Figure 2.2: LoRA reparametrization from [32]

up the process.

With the low-rank matrices effectively capturing the specific knowledge needed for a new task, the model can adapt to new data without modifying its core parameters by freezing the original model. Making the application of LoRA in a “plug-and-play” fashion. This flexibility is particularly beneficial for applications where models need to be adapted frequently for different tasks.

LoRAs can be specifically useful in this project, especially for the task of generating different emotions given the player’s input photo with Stable Diffusion. As a different LoRA can be trained for a specific emotion, without making any changes to the original model. With the help of LoRAs, the fine-tuning for adaptation will be faster and much easier while ensuring that the generated images are tailored to the specific emotional needs of the user, thereby supporting a more engaging and effective therapeutic process.

2.5.4 ControlNet

ControlNet is a neural network architecture designed for diffusion models. The technique, as its name suggests, provides enhanced control over the image generation processes. ControlNet enables the conditioning of diffusion models with additional input information, such as edges, poses, depth maps, and other structural cues, which allows for more precise and detailed generation of images according to user specifications [33].

ControlNet operates by taking a pre-trained diffusion model and augmenting it with a control network that can process additional conditioning inputs. These inputs can be edge maps, pose estimations, semantic segmentation maps, or other structural representations that provide a more detailed context for the generation process. By incorporating these conditions, ControlNet can manipulate the underlying diffusion model to adhere more closely to the desired output [33].

ControlNet expands the capabilities of diffusion models by enabling more precise control over image generation, which is what the somewhat unpredictable diffusion model needed. This technique has the potential to shine in this project as the desired expression can be encoded by a ControlNet. By applying the ControlNet to the player's input image, it is possible to generate the image with the desired facial expression.

2.6 Serverless Computing

Serverless computing is a cloud computing execution model where the cloud provider dynamically manages the allocation and provisioning of servers. In a serverless model, developers can build and run applications without having to manage the infrastructure, as the cloud provider takes care of the server management, capacity planning, scaling, and maintenance [34]. It might be a bit confusing at first, but the term “serverless”, instead of meaning no server is needed, implies that developers do not need to be concerned with server management tasks. Serverless services and frameworks enable developers to focus on writing code and deploying applications efficiently, often leading to faster development cycles and reduced operational costs.

2.6.1 Mechanism of Serverless Computing

The most common form of serverless computing is Function-as-a-Service (FaaS). In FaaS, individual functions are deployed and executed independently. Each function is stateless and triggered by events such as HTTP requests, database changes, or file uploads [35]. Examples of popular FaaS platforms include AWS Lambda, Google Cloud Functions and Alibaba Functions, which this project is using.

2.6.2 Advantages

- **Cost Efficiency:** Serverless computing allows users to only pay for the compute time they consume, rather than provisioning and paying for fixed amounts of server capacity [34]. This pay-as-you-go pricing model reduces operational costs, especially for this project, where the server is inactive most of the time.
- **Reduced Operational Complexity:** Serverless abstracts the underlying infrastructure, allowing developers to focus on writing code and deploying applications without worrying about server management [36].

2.6.3 Disadvantage(s)

- One of the significant issues with the serverless is the “cold start” latency which typically occurs when functions that haven't been used recently are invoked, leading to a delay as the server initializes the environment [37]. This can impact performance for time-sensitive applications.

2.6.4 Conclusion

The serverless computing model is ideal for this project, especially for hosting large deep learning models such as Stable Diffusion. The pay-as-you-go nature makes it really cost-effective, especially for a GPU server.

2.7 Unity

Unity is a powerful and widely-used game development platform that enables developers to create interactive 2D and 3D content for a variety of platforms, including desktop, mobile, console, and virtual reality (VR) devices. Developed by Unity Technologies and first released in 2005, Unity has become one of the most popular game engines in the industry due to its flexibility, robust feature set, and supportive community [38]. It is equipped with a wide range of features that make it suitable for developers at all levels, including beginners like the author of this project:

1. **Cross-Platform Development:** Unity allows developers to build games and applications for multiple platforms from a single codebase. This includes major operating systems like Windows, macOS, and Linux, as well as gaming consoles, mobile devices, and AR/VR platforms [39]. This cross-platform capability significantly reduces development time and effort, enabling developers to reach a broader audience with their applications.
2. **Lively Community and Comprehensive Asset Store:** The Unity Asset Store features a variety of plugins and tools that extend Unity's capabilities created by the community, allowing for easy integration of third-party services and features. Saving time and effort for developers and make them focus on more creative or project-specific work. The vast popularity also makes tutorials easier to find online, making the platform easy to start with.
3. **C# and Visual Scripting:** Unity uses C# as its primary programming language, which is known for its ease of use and versatility. In addition, Unity supports visual scripting through tools like Bolt, allowing non-programmers to create gameplay mechanics and interactions without writing code, which is very beginner-friendly.
4. **Integrated Development Environment (IDE):** Unity comes with a fully integrated development environment that includes a scene editor, asset management tools, physics engine, animation editor, and more. This comprehensive toolset enables developers to design, develop, and debug their applications all within a single platform [39].

Many famous games were developed using Unity, particularly in the 2D genre, which is relevant to this project. Notable examples include *Ori And The Blind Forest* [40], *Cuphead* [41], *Rimworld* [42], etc. All of those demonstrate Unity's capability in creating visually stunning and mechanically robust 2D games.

To sum up, Unity is a highly promising tool for 2D game development. Its integrated features, flexibility, and extensive community support make it an excellent choice for beginners, providing a user-friendly and efficient environment that allows developers to focus on the creative aspects of their projects.

Chapter 3

Case Study: *The Last of Us Part II*

Many narrative-driven games on the market try to put the players in a virtual moral dilemma and utilize player agency, choice-based narrative and branching to make the players reflect on their choices, did they make the “right” one? The game *Detroit: Becoming Human* [43] is such an example. By plotting an interesting story and a logical future setting, it creates a real-like game world, makes the player feel the conflicts between the “androids” and humans, making the players to think about the boarder theme of the game: what does being a human actually mean? The game also reflects real-world issues, such as racism and civil rights. The game is widely praised for its storytelling, graphics, and the depth of its branching narrative, offering a thought-provoking experience, particularly for those interested in narrative-driven games.

However, narrative-driven gameplay is challenging as a balance should be made between narrative and player agency, and the motivations and story development must be logical enough to keep the world real and immersive. This chapter examines *The Last of Us Part II* as a case study to understand the complexities and potential pitfalls in storytelling in narrative-driven games. By analyzing the game’s controversial reception, lessons can be drawn to avoid similar missteps in future story designs.

The Last of Us Part II (TLOU2), developed by Naughty Dog and released in 2020, is a narrative-driven game that explores themes of violence, revenge, and forgiveness. The story follows two main characters, Ellie and Abby, who are both driven by personal loss and the desire for vengeance. The game aims to depict the devastating consequences of revenge and the potential for breaking the cycle of violence through forgiveness [14].

3.1 Narrative and Thematic Ambitions

TLOU2 sets out to challenge players’ perceptions of revenge and forgiveness by presenting a narrative where characters are caught in a cycle of violence. Ellie’s journey is a quest for revenge against Abby, who killed her father figure, Joel. Throughout the game, players witness the destructive consequences of Ellie’s obsession with re-

venge, not only on her own psyche but also on the people around her [44]. The narrative attempts to convey that revenge perpetuates violence and suffering, while forgiveness has the potential to break this destructive cycle.

3.2 Criticisms of Moral Lessons

Despite its narrative ambitions, TLOU2 faced significant criticism for what many players saw as an unrealistic and forced moral lesson regarding revenge and forgiveness. After spending most of the game pursuing vengeance, Ellie's ultimate decision to forgive Abby felt abrupt and unsatisfying to some players [45]. This resolution seemed to undermine the emotional stakes that had been built throughout the game, as players expected a more nuanced or cathartic conclusion to Ellie's journey.

The game's insistence on forgiveness as the only path to breaking the cycle of violence was also seen as overly simplistic by many players. Critics argued that the narrative failed to fully explore the complexities of revenge and forgiveness, particularly in a world where survival often hinges on violence. The abrupt shift from revenge to forgiveness left some players feeling disconnected from the emotional journey they had been part of for most of the game.

3.3 Forgiveness vs. Revenge

The tension between forgiveness and revenge is central to *The Last of Us Part II* (TLOU2), sparking significant debate among players and critics [46]. Many players expressed dissatisfaction with the way the narrative seemed to abruptly push the theme of forgiveness. For these players, the storyline's resolution, where Ellie chooses to forgive Abby instead of exacting revenge, felt forced and unrealistic given the brutal world setting and the extent of Ellie's losses and suffering throughout the game [45]. This perceived lack of narrative balance has led to criticism that the game imposes a singular moral viewpoint—one that advocates for forgiveness without fully exploring the complexities of vengeance in a world where survival often depends on it.

On the other hand, some players and critics appreciated the game's attempt to deliver a message about the futility of revenge. These players found the story's conclusion, where Ellie ultimately chooses to break the cycle of violence, to be a powerful statement on the potential for empathy and redemption, even amidst overwhelming grief and anger [47]. For these players, the game's narrative choices reflect a deeper understanding of the emotional costs of revenge and the possibility of healing through forgiveness.

The debate reveals a significant divide in how audiences perceive the moral lessons of TLOU2. While the game attempts to portray the destructive nature of revenge and suggest that forgiveness can lead to emotional peace, its heavy-handed approach to

this theme can significantly damage player immersion and overall game experience. By disregarding player expectations and imposing a seemingly unrealistic narrative of forgiveness, especially in a world marked by brutality and loss, the game risks alienating its audience. Players may feel that their emotional investment in the characters and story is undermined by a forced resolution that does not align with the intense, complex emotions depicted throughout the game. This disconnection can lead to resistance against the message the creators intended to convey, as players feel that their agency and emotional journey are not respected [45].

3.4 Healing as the Key to Breaking the Cycle

The author of this article believes that breaking the cycle of violence does not hinge solely on forgiving and giving up revenge. The moral dilemma of whether to pick forgiving or revenge is a complex topic and neither approach is perfect for all situations. TLOU2 was trapped in this dilemma and narrative pitfall, causing heated debate between its supporters and haters.

The author believes that instead of being trapped in debates between those two, for the educational purpose of this project, healing and reconstruction are far more crucial elements in breaking the cycle of violence. The process of overcoming trauma involves much more than a simple moral decision; it requires self-repair, emotional growth, and a commitment to rebuilding oneself and one's community. It is also essential for individuals who have not experienced such trauma to understand these processes, so they can better support victims in their journeys toward healing and offer empathy and understanding.

3.5 Conclusion

The case study of TLOU2 provides a solid insight for this project by highlighting both the potential and the pitfalls of tackling complex moral themes in narrative-driven games. TLOU2's narrative ambitions to explore the cycle of violence and the possibility of breaking it through forgiveness demonstrate the powerful role that storytelling can play in engaging players with deep, emotionally resonant themes. The game attempts to challenge players' feelings and encourage reflection on the consequences of revenge, illustrating how narrative-driven games can serve as a medium for exploring difficult moral questions.

However, the mixed reception of TLOU2's narrative also underscores the challenges inherent in such storytelling. While TLOU2 aimed to explore the cycle of violence and the possibility of breaking it through forgiveness, its focus on a forced moral resolution alienated some players and hindered the narrative's emotional impact. The game's tendency to frame the conflict as a binary choice between revenge and forgiveness detracted from the broader and more critical theme of healing after trauma.

For this project, the focus shifts away from forcing players into a strict moral dilemma, setting the stage in an environment where revenge is already served lawfully so that the player can put more focus on the promoted healing and empathy theme. With an open ending and by presenting what the world has become because of the player's choices, the game provides a more friendly approach that respects every player's choice.

Ultimately, the lessons learned from TLOU2 will guide this project in crafting a story that respects player agency, emotional complexity, and the importance of healing as the true key to breaking the cycle of violence.

Chapter 4

Story Design

Story design is a crucial component of this project. By providing a logical, emotional and thought-provoking story, an immersive game world will be built upon it. Utilizing the empathy of the players, the players are expected to experience a “virtual trauma” in the game, which serves as a solid foundation for the SAT exercises aiming at fostering empathy, understanding, and emotional growth. After the SAT exercises, the story also provides a light-weight choice-based narrative section, innovatively in the style of Twitter posts. This not only tests the result of SAT protocol but will also make the player reflect on their choices.

The purpose of this chapter is to give an overview of the storyline, narrative framework, character development, specially designed interactive storytelling and how they align with the project’s educational and therapeutic goals.

4.1 Storyline Overview

The story focuses on two main characters: the protagonist, controlled by the player, and his close friend Ian. Both children live in a somewhat isolated village and share a deep curiosity about the outside world. Despite their eagerness to explore beyond their village, the majority of the adults in their community are wary of the outside due to past experiences with greed and malice. In fact, their ancestors settled in this village to escape what they saw as a corrupt and hostile world.

One day, a man named Morris – a figure of greed and malice – discovers the village. Seeing the potential of the land, he attempts to claim it for himself and makes an offer to the villagers. However, his proposal is firmly rejected, as the villagers have no interest in aligning with his kind. Feeling humiliated by the rejection, Morris decides to take the land by force. Later that night, he invades the village and commits a brutal genocide, justifying his actions as a mission to rid the world of what he views as uncivilized filth.

The two main characters – Ian and the protagonist, survived by accident somehow. Traumatized by the loss of his village and family, Ian gets consumed by anger and

a desire for revenge. Through relentless effort, he becomes a lawyer and eventually brings Morris to justice. This earns him widespread acclaim on the Internet, especially from those who have also suffered at Morris's hands.

However, Ian's unresolved childhood trauma leads him down a darker path. His original curiosity about the outside world has turned into a deep distrust, and he begins to lose control. With his rising influence, Ian starts calling for justice against more individuals he deems guilty. Over time, his once-righteous campaign transforms into a dangerous witch hunt, where innocent people are increasingly caught up alongside the guilty, blurring the lines between justice and vengeance, forming a cycle of violence.

On the other hand, the protagonist, as well as the player, will go through the SAT exercises after the genocide. Guided by the principles of the Self-Attachment Technique (SAT) and the teachings he received from his family, the protagonist seeks healing and understanding rather than revenge. He becomes a community volunteer, focusing on supporting those who, like him, have experienced profound loss.

In the game's climax, the player faces a critical choice: to join Ian in his vengeful crusade or to stand up against the cycle of violence and advocate for a more compassionate approach. Through posting on an online social media called "Tweetz", the player can either choose to speak for Ian or advocate peaceful interactions. The game's ending is open, but the process is determined by the player's choices. The choices and those tweets reflect the broader theme of breaking or continuing the cycle of violence, challenging players to consider their values and the impact of their actions.

4.2 Character Development

4.2.1 Ian

Ian's character development is a central pillar of the game's narrative, serving as a powerful exploration of how childhood trauma can shape an individual's path. His journey is marked by significant changes in personality, motivations, and actions, reflecting the complex interplay between his internal struggles and external circumstances.

Early Characterization

At the beginning of the game, Ian is depicted as a curious and adventurous 13-year-old boy, eager to explore the world beyond his isolated village. He is frustrated with the village's conservative mindset and the adults' constant warnings about the dangers of the outside world. This initial characterization paints Ian as an idealistic youth, full of mind and exploration, but feels limited by the constraints. It also gives a hint of Ian's escaping from the village.

Escaping from the village

When the village's attention was drawn to Morris, Ian saw his chance and finally made up his mind to escape. However, as a 13-year-old, his impulsive decision quickly turned to regret once he found himself alone outside the village. The reality of being away from his family and the safety of his home struck him hard, making him realize the importance of the connections he had taken for granted. This incident marks a significant turning point in Ian's attitude towards the village and the family he once disliked and wanted to escape. It sets the stage for his future character development, especially after he returns to the village and discovers the devastating aftermath of the genocide.

The Genocide

The genocide of his village marks a dramatic turning point in Ian's life. Witnessing the destruction of his home and the brutal loss of his family fills him with a deep sense of grief, fear, and self-blame. Initially overwhelmed by these emotions, Ian soon finds them transforming into intense hatred and a powerful urge for revenge when he sees Morris and his followers celebrating their horrific deeds. This traumatic event shatters Ian's youthful innocence and replaces his curiosity about the world with a profound distrust of outsiders. This dramatic transformation aligns with the theories proposed in [2], highlighting the psychological impact of mass violence and potentially fostering greater empathy from the player. It also provides the emotional foundation for Ian's eventual descent into a cycle of revenge.

Summary

Ian's character development is a complex journey from innocence to vengeance, mirroring the core themes of the game, such as the cycle of violence, the moral complexities of justice, and the power of empathy. His descent into a cycle of violence serves as a vivid illustration of the consequences of unchecked anger and the desire for retribution. Ian's story is a poignant reminder of the importance of addressing trauma and seeking healing, rather than allowing pain and anger to fuel further conflict. This setting is beneficial for making the player reflect on their potential roles in conflict resolution and promote peaceful interactions. The gradual and realistic development of Ian's character contributes to creating a more believable virtual world, enhancing the players' ability to empathize with his experiences and choices, and adding more depth to the storyline while making the players reflect on how they would react themselves in this situation.

4.2.2 The Protagonist

The protagonist's character development is intentionally less defined to allow the player to project their own thoughts and emotions onto the character. At the start of the story, the protagonist is a child turning 13, sharing a deep curiosity about the outside world with his friend Ian. He lives in a small, isolated village and has

a close-knit, loving family, consisting of his parents and a younger sister. The protagonist's life is depicted as happy and peaceful, filled with the innocence and joy of childhood, and sheltered from the darker aspects of the world beyond the village.

Deeply affected by the loss and horror of the genocide, the protagonist seeks healing and understanding rather than revenge. He follows the principles of the Self-Attachment Technique (SAT) and the values his family taught him, dedicating himself to fostering compassion and empathy in a world filled with violence.

As the player controls the protagonist, they are invited to experience this journey of healing firsthand. The protagonist's character is shaped by the choices the player makes in the game, including key decisions about whether to pursue revenge or seek to break the cycle of violence. This interactive element allows players to explore different aspects of the protagonist's potential responses to trauma, ultimately guiding the character towards an outcome that reflects the player's own values and beliefs.

4.3 Underlying Themes & Designs

The story generally includes several main themes that are expected to serve the goal of the game

4.3.1 Genocide

One of the central themes of the project is the trauma caused by genocide, which serves as a key narrative driver in the game. The design of the genocide in the game is not merely an act of mass violence but is framed through the lens of dehumanization – where one side perceives and treats the other as sub-human. In the game, this is typically reflected in Morris's constant calling the people in the village "rats", portraying them as sub-human beings who refuse to be civilized. Morris's constant degradation of the village people serves to strip them of their humanity, making it easier for him and his followers to commit atrocities without guilt or remorse. He justifies his violent actions by claiming he is "doing the world a favour to cleanse the filth," thus framing his genocide as a noble mission rather than a crime against humanity.

This design choice mirrors real-world instances of genocide, where perpetrators often employ ideological, racial, or cultural superiority as a justification for their crimes. By showing how the moral decay that stems from dehumanization leads to profound trauma for survivors, the narrative emphasizes the psychological impact of such violence and the horrors of using ideological beliefs to rationalize atrocities. Presenting genocide in this manner also sheds light on the emotional devastation caused by dehumanization, illustrating the long-lasting effects on victims.

Through this portrayal, the game aims to create a virtual trauma experience for players who may not have personally encountered such extreme violence. By immersing

players in a story that mirrors the psychological and emotional devastation of genocide, the game provides an opportunity to empathize with victims and understand the broader ethical implications of dehumanization. This virtual experience serves as an educational tool, fostering empathy and raising awareness of the complex factors that lead to mass violence. Players are not just passive observers but are encouraged to reflect on how such atrocities unfold, building a deeper understanding of the psychological and societal consequences of violence.

4.3.2 The Cycle of Violence

The cycle of violence [1] is a central theme in the game, exploring how violence breeds more violence, leading to a never-ending loop of retaliation and suffering. Through the development of the character Ian, the game demonstrates the destructive power of hatred and revenge. Ian's quest for vengeance following the genocide that devastated his village exemplifies how trauma can perpetuate further violence, which follows [2]. In contrast, the protagonist's journey toward healing represents the possibility of breaking this cycle through empathy and understanding. The game challenges players to consider how they might break free from destructive patterns in their own lives and communities.

4.3.3 Revenge vs. Forgiveness

This theme is carefully dealt with in the story design process. A common narrative pitfall, which is shown in the case study of TLOU2, is equating the idea of breaking the cycle of violence solely with forgiveness. While relinquishing revenge can indeed be a powerful means of breaking the cycle, the author argues that in some situations, such as facing a character as malevolent as Morris in the game, this approach may be impractical.

Breaking the cycle of violence does not inherently require choosing forgiveness over revenge. Instead, the focus should be placed on the importance of healing after trauma. In this game, revenge serves as a natural response for the characters, but it is through emotional healing—not merely the abandonment of revenge—that the cycle can be truly broken. The story design emphasizes that the path to breaking the cycle involves fostering self-awareness, healing, and personal growth, rather than presenting forgiveness as the only solution.

In the game, Ian seeks his revenge through lawful means, which allows the narrative to focus on the broader implications of the cycle of violence without becoming trapped in debates about the morality of revenge. This approach encourages players to reflect on the consequences of both vengeance and forgiveness, making their choices more meaningful within the context of the game. The player must decide whether to support Ian's pursuit of justice or advocate for breaking the cycle of violence, thereby directly engaging with the game's central themes and reflecting on their personal values.

4.3.4 Healing & Empathy

Healing and empathy are core themes that are emphasized through the integration of SAT exercises within the game. These exercises encourage players to engage in self-reflection and emotional management, fostering a sense of compassion for themselves and others. By guiding players through a process of “re-parenting” their childhood self, the game aims to promote understanding and emotional resilience, highlighting the importance of empathy in overcoming trauma and building healthier relationships.

4.3.5 Justice & Morality

The game also explores a bit about the theme of justice and morality, questioning what true justice looks like and how it should be achieved. Ian’s transformation from a victim seeking justice to a figure leading a potentially dangerous movement raises important questions about the moral boundaries of justice. The game encourages players to consider the fine line between justice and revenge, and how moral choices can have far-reaching consequences for individuals and society.

4.4 Interactive Storytelling

Interactive storytelling is a key feature of this game, designed to enhance player immersion and engagement. By allowing players to actively participate in the narrative, the game creates a more compelling and emotionally resonant experience. This section outlines the methods used to create a richer, more interactive game world.

4.4.1 Interaction with NPCs

In the first set of scenes, typically before the genocide, players are given the freedom to explore the village and interact with its various NPCs. This approach offers a more engaging way for players to gather information, compared to simply presenting text-based exposition. The interactions are entirely optional, allowing the game to progress without them, but they add significant depth to the story for those who choose to engage.

For instance, interacting with the village bookkeeper provides insights into the village’s history, helping players understand the underlying tensions between the villagers and outsiders. This interaction not only enriches the narrative but also subtly guides the player towards understanding the broader context of the conflict.

The NPC interactions also serve as a natural way to guide players through the game. For instance, when the player first steps outside the house, they might not immediately know where to go to advance the storyline. However, by choosing to interact with NPCs, players can receive hints about what to do next. An NPC might mention

that something unusual is happening near the village gate, directing the player toward the scene’s objective without breaking immersion with overt instructions.

Additionally, a side quest involving the protagonist’s sick little sister adds an emotional layer to the gameplay. She asks the player to find a beautiful flower for her, a request that players can choose to fulfill or ignore. If the player finds and gives the flower to the sister, it becomes a detail later in the story, as the flower is shown dropped on the ground during the night of the genocide. This sub-quest enhances the narrative by providing a tangible connection between the player’s actions and the unfolding tragedy, making the game world feel more alive and emotionally impactful.

4.4.2 Scripted Events and Cutscenes

Cutscenes are used a lot in the game. While these are not always interactive, they are strategically placed to enhance the storytelling experience by providing “cinematic” moments or deeper insight into the game’s narrative.

4.4.3 Environmental Storytelling

The game employs environmental storytelling to convey narrative elements through the visual and atmospheric details of the surroundings, catering to players who are adept at interpreting information from the environment. As shown in Figure 4.1, in the initial peaceful village scene, for example, vibrant, “moving” flowers and grasses are present in the background, contributing to a serene and joyful atmosphere. These elements subtly set the tone, immersing the player in the happiness of the village. However, after the genocide, these flowers are gone, replaced by a stark and desolate landscape that reflects the shift to a somber and serious mood.



Figure 4.1: Part of Peaceful Village Scene

As the player navigates the village in the aftermath of the genocide, the environment tells its own story of the tragedy. Figure 4.2 illustrates this with part of the scene. Dead bodies are pierced and stripped off, each depicting a different and brutal end, revealing the cruelty of Morris and his men. Some of these bodies belong to characters the player may have interacted with earlier in the game, making the scenes more emotionally impactful.



Figure 4.2: Part of Aftermath Scene

The aftermath environment also provides clues to the motivations and actions of the aggressors. For instance, the mayor's house is shown to be the most heavily damaged, suggesting a targeted attack. When combined with the earlier narrative that the mayor refused Morris's offer, players can deduce that Morris has a particular hatred for the mayor and his family.

This use of environmental storytelling not only deepens the player's emotional engagement but also enhances the richness of the game's narrative, encouraging players to piece together the story through observation and interpretation.

4.4.4 Choice-based events

Choice-based events play a crucial role in shaping the narrative, especially in the concluding "Tweetz" scene. Although earlier gameplay includes some choice-driven dialogues, these choices do not significantly impact the overall story. However, the "Tweetz" scene is designed to emphasise the player's influence on the narrative's direction.

Figure 4.3 shows what the scene looks like. In the "Tweetz" scene, players can choose to post at certain points, showing their attitude towards Ian's approach. Much like real-life social media, the "Tweetz" platform uses metrics to measure the player's influence and the reception of their posts. The number of followers indicates the



Figure 4.3: Part of Tweetz Scene

player's influence and whether the public is aligning with their viewpoint. The number of likes, comments, and retweets reflects the popularity or controversy of a tweet, providing insight into the public's reaction.

This mechanic is intertwined with environmental storytelling, allowing observant players to glean additional narrative details from the metrics. For example, a post with many likes might suggest widespread approval of an idea, while a post with few likes but many comments could indicate controversy or debate. The scene also focuses on several NPCs, tracking their replies and posts throughout. This builds a more realistic society, adds immersion, and makes it more interesting to see how their posts evolve.

There are three stages of player choices:

- **First Stage:** This stage occurs right after Morris is taken down, and the crowd is enthusiastic about Ian's bravery. The player can advocate for peaceful interactions and healing, praise Ian, or call for more cautious revenge. Praising Ian gains the most followers since he is seen as a hero at this point, while the other two choices seem less "appropriate" at this stage. However, this choice is a warm-up and does not impact the storyline.
- **Second Stage:** By this stage, Ian has become more extreme, and his supporters are showing signs of craziness. The player can choose either to continue praising Ian or focus on healing rather than revenge. If the player chooses the former, they lead to a chaotic world where Ian becomes more obsessed with revenge. Choosing the latter option makes the hidden choice in stage three available.

- **Third Stage:** This stage occurs when things begin to get out of control, as the "justice movement" turns into a "witch hunt." The player can choose to continue supporting revenge or call for careful justice. Both choices lead to the same ending, as they do not break the cycle of violence. Only if the player chooses to focus on healing in stage two can they advocate for breaking the cycle of violence in the third stage. This choice may not show immediate effects but persuades a previously Ian-supporting user to adopt the idea of breaking the cycle. In the ending scene, this user will post about breaking the cycle of violence, and the player can see a significant increase in followers.

This approach not only deepens player engagement but also effectively immerses them in the emotional landscape of the narrative. By simulating complex moral dilemmas and consequences, the game mirrors the psychological impact of real-world trauma, making the player feel the weight of their choices. This immersive storytelling technique is highly beneficial for the Self-Attachment Technique (SAT) to work effectively, as it encourages players to confront and process intense emotions within a controlled environment.

4.5 Summary

Through careful design and the use of multiple techniques and mechanisms in storytelling, the story itself is robust for providing an immersive world with realistic interactions. This makes the trauma of the genocide more real to the players, which serves as a solid foundation for the SAT exercises to be effective.

Chapter 5

SAT Integration

As discussed in the previous chapters, the SAT integration is one of the core parts of this project. The SAT exercises have been shown to have great potential in regulating emotions and promoting compassion[8], reducing the level of anxiety and depression [12], and improving overall wellbeing. [13].

Having the SAT exercises integrated right after the genocide in the game, this project seeks to make the players empathize with the main character in the game, and utilize the benefits of SAT exercises to help regulate their emotions after experiencing such intense “virtual trauma”, help them develop self-compassion and understand the root of the conflict. It is hoped that through the exercises, the player can choose healing and peace over revenge, eventually breaking the cycle of violence.

It is worth noting that the SAT exercises in the game, which are slightly different to the previous use cases, mainly focus on the “virtual trauma” the player experienced in the gameplay, rather than his/her own past pain. This is specially designed so that it can, instead of breaking the immersion of the game, maximize the players’ empathy and compassion so that they can focus more on the genocide that just happened in the game.

5.1 SAT protocols used

In practice, the SAT protocol being used in this project is a selection from [12], as these exercises have already proven useful in regulating emotions and fostering inner peace. The detailed protocols include:

1. In a quiet place, look at the happy and unhappy childhood photos and recall positive and negative childhood memories and early relationships in the family.
2. Image being with the happy child with the help of the childhood photo, then imagine embracing the child, then playing a game with the child and finally dancing with the child.
3. Imagine being with the unhappy child, picture embracing and comforting the child until the child finally becomes happy.

4. Sing a favourite love song that shows love and affection to the child while looking at the happy childhood photo.
5. Express the love and care for the child out loud. While looking at the picture of the happy childhood photo, make a vow to care for the child as a loving parent.
6. Project the positive and negative emotions to the corresponding photos and create an emotional world for both. For example, positive emotions can be pictured as a new home that provides a safe haven, while negative emotions can be seen as an unrestored basement where the child is trapped in.
7. Maintain a loving relationship with the child and maximise the positive emotions by singing, shaking the body and dancing while looking at the happy child photo.
8. Create a bond with nature. Making nature an attachment of the child can help to increase joy and reduce negative emotions.
9. Picture being near the unhappy child by looking at the unhappy photo. Loudly reassure the child and give a face/neck/head self-massage until feeling calmed and comforted.
10. Recall a painful childhood episode, associate the face of the child in the past experience with the unhappy childhood photo. Imagine the adult self intervening like a good parent by approaching the child quickly, loudly reassuring the child that they are safe and standing up with a loud voice to the perpetrator. Then imaginatively cuddle the child. Repeat the process until feeling comforted, acquiring mastery over the trauma.

These exercises provide a basic framework that is proven to be useful in fostering inner peace and regulating emotions. It is worth noting that the original exercises are done with the help of the VR technique, and this project has made several changes to make the exercises fit in the narrative-driven gameplay.

5.2 “Gamified” SAT Exercises

As discussed previously, while players are still using their own childhood photos and avatars, they are supposed to complete the SAT exercises with the genocide as their childhood trauma, so that they can develop more understanding of this pain, which contributes to building a more peaceful society that help break the cycle of violence. Several exercises are thus “gamified” to reflect this change, this typically includes:

1. While playing with the childhood self in exercise 2 listed above, the child will challenge the player with a hide-and-seek game. Which is a callback of the previous storytelling mentioning the protagonist knows a hidden door under his bed that nobody else knows and he always wins the game because of this. This change not only exemplifies the original SAT protocol’s requirement of

playing with the child but also builds up the story, making the story more holistic.

2. For building the emotional world in exercise 6 listed above, the house the protagonist once lived in serves as a safe haven for positive emotions while the house after genocide acts as the one with negative emotions. This setting fits more in the game and saves the players from having to imagine themselves.
3. For building a bond with nature in exercise 8, considering the player may not be available to go out while playing, the game sets the scene to happen in the peaceful village the protagonist once lived. In addition, it provides several scenery images generated by Stable Diffusion that feature the village’s landscape, such as a sakura tree, a beautiful lake and more. This, hopefully, will also make the world seem more realistic to the player.
4. For overcoming past pain in exercise 10, the exercise focuses specifically on the genocide trauma. Guided by two cutscenes featuring the adult protagonist caring for the child by comforting him and driving away Morris, the player can hopefully get a taste of what the exercise is about and be more focused on the topic of genocide and breaking the cycle of violence.

In summary, the “gamified” SAT exercises, while retaining the core idea of those previously proven to be effective, fit better in the narrative framework of the game and serve as a promising attempt to make the SAT more focused on the topic of genocide and breaking the cycle of violence.

Chapter 6

Facial Expression Changer

As shown in the chapter about SAT exercises, it is clear that the childhood photo of the player with different facial expressions plays an important role in the process of SAT. In the typical use cases, VR equipment was provided to support the SAT process. By manually setting up the virtual avatar of the player's childhood self, the VR environment makes it convenient to visualize the child's avatar with different emotions, smoothening the SAT process. However, this is not practical for this project, as the game is set to be an accessible therapeutic and educational tool, and the VR environment will higher the requirements for this.

Having the players manually upload their childhood photos is a valid plan, but it is not realistic for the player to have enough clear childhood photos at hand, let alone all the facial expressions needed, namely:

1. Sad
2. Angry
3. Disgusted
4. Fearful
5. Surprised
6. Neutral
7. Happy

This poses a barrier for the SAT integration in the game, as the exercises can be hard for the players to proceed purely based on imagination and without any material help.

The facial expression changer is developed to address this problem and help the player have a smoother SAT experience. Powered by Stable Diffusion [29] and ComfyUI [31] in the backend, it allows the player to upload only one (childhood) photo, regardless of the expression, and create all the photos needed for the later SAT exercises by simply clicking one button. To make the image generated fit more to the

general art-style of the game, a model specifically fine-tuned for pixel art [48] is used. This pixel art style also helps in the situation where the expression change to the real photo can be a little faulty (such as blurry or distorted) to human eyes.

The project also leverages the power of the cloud GPU server to make it possible for the large Stable Diffusion model to run on almost every machine, lowering the requirements of the computer resources, especially the GPU, making the game more accessible to the people in need.

In this chapter, several ComfyUI workflows designed by the author to complete the task are explained and compared. The chapter will also touch upon how the cloud server is set up to reduce the cost.

6.1 Workflow Design

Several workflows were designed and compared during the development of the expression changer for better user experiences.

6.1.1 Basic Image-to-Image workflow

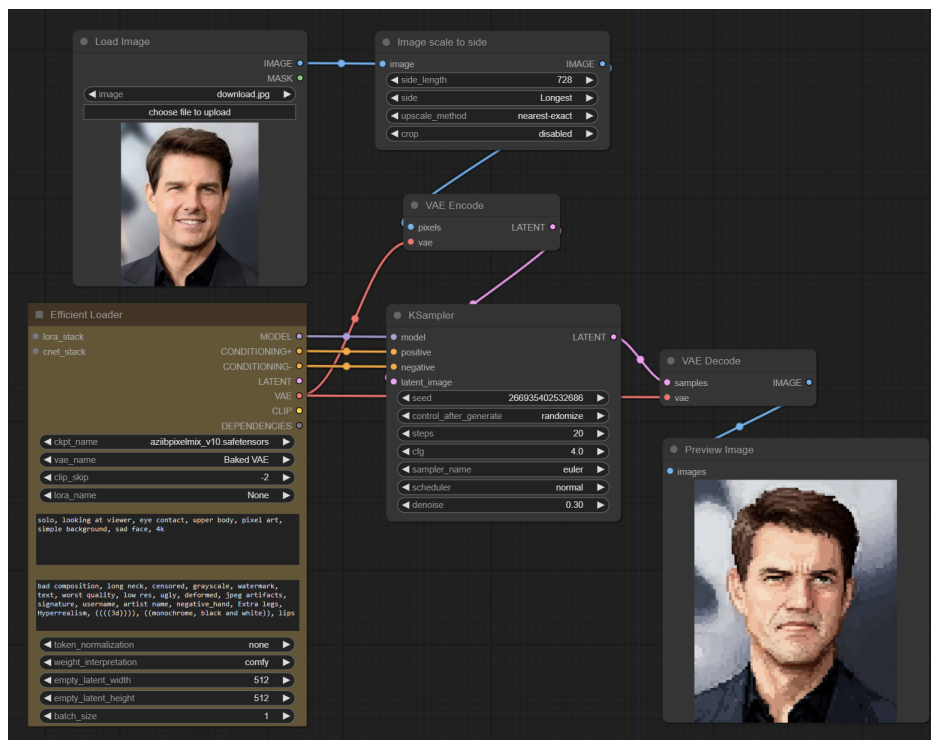


Figure 6.1: Basic image-to-image workflow

Figure 6.1 shows a basic image-to-image workflow that is capable of changing the input photo's expression in ComfyUI, and Figure 6.2 is a simplified version of it. The

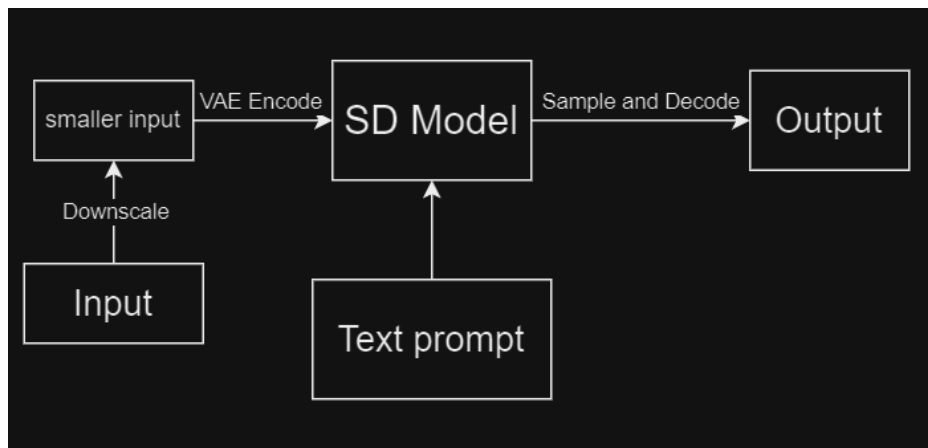


Figure 6.2: Simplified Basic image-to-image workflow

workflow works as follows: It first takes an input image, and it downscales the image so that the longest edge of the input is 728 pixels long. The downscale typically makes it faster for the Stable Diffusion model to process the input image, and the 728 is a number gained from experiments that is generally good for all sorts of inputs, regardless of the aspect ratio. The input is then encoded by the VAE encoder of the model (section in red in Figure 2.1) into the latent space. A random noise is then sampled for the model to generate, conditioning with the text prompt and the input image, the generated image generally captures the desired feature of a sad face in Figure 6.1 given the text prompt.

The denoise parameter in the KSampler node typically sets the similarity between the generated image and the input photo. The lower the value is the output will be more similar to the input, with 1 being the output completely different from the input and 0 means the two are exactly the same. The value 0.3 is gained through experiments and experiences, and proven to be working in most cases.

The CFG value in K-Samplers stands for Classifier-Free Guidance scale. It is a parameter used to control the strength of guidance during the image generation process. When the CFG value is low (typically below 3.0), the model combines the guided and unconditioned predictions almost equally. This setup often results in more diverse and sometimes unexpected outputs that are not as tightly constrained to the prompt. While a high CFG value makes the model adhere more closely to the prompt.

Evaluation

Advantages:

- The workflow is very simple while being capable of doing the task of expression change.
- The workflow is very fast for each image generation, typically 3 seconds per

image in an environment with a 3080Ti laptop GPU.

- Nice pixel-style art.

Disadvantages:

- The generation is less controlled, and the quality of the generation may vary.
- Similarly, due to less control the faces may change so the generated image sometimes does not look like the original person.
- While this workflow is capable of generating simple expressions, such as sad and happy, more subtle ones, such as disgusted and fearful, are hard to get through simple prompting, as shown in Figure 6.3, and the cost of looking for good prompting is too high yet not stable.



Figure 6.3: Different expressions generated by the basic workflow

6.1.2 Final Workflow

Figure 6.4 shows the final workflow the project is using in ComfyUI, and Figure 6.5 shows a simplified version of it. The workflow comprises three parts: the first part uses a pre-trained stable diffusion model that is fine-tuned for high-res real-like images from [49] to do the expression change task first. The Face Detailer node, which is an extension for ComfyUI in the ComfyUI-Impact-Pack [50], is basically a KSampler but utilizes computer vision models, such as YOLO (You Only Look Once) model for face detection and SAM (Segmentation Any Model) for segmenting out the face for detail changes. As can be observed in Figure 6.4, the Face Detailer outputs the photo with only the face changed according to the prompt, in this case, a happy/smiling version of the original one.

The modified photo is then upscaled, which is a preparation for the later pixel-style transfer. The photo is then encoded to latent space using the second Stable Diffusion model, which is the pixel-style model used for the previous workflow. Through the control of another set of LoRAs and prompts, the control is reinforced for the pixel-style output. This part also serves as a safety check stage, as the output from the first stage may not always be as stable and there might be some artefacts or misplacement in the Face Detailer's output. By adding the pixel-style filter and re-emphasizing the

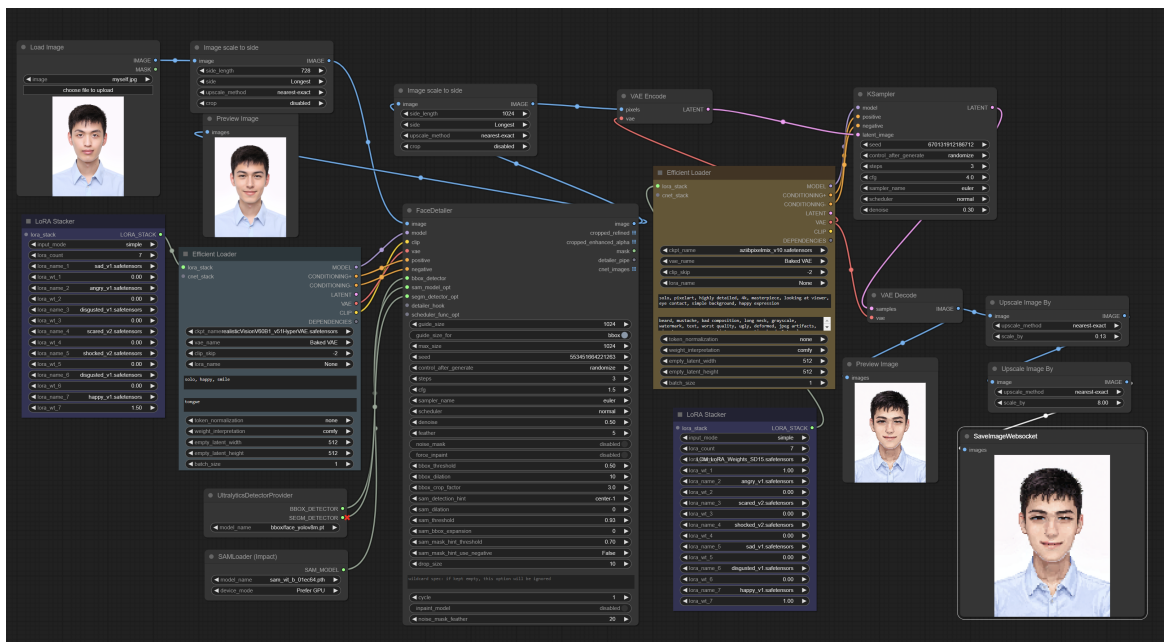


Figure 6.4: Final expression change workflow

desired features, the second part of this workflow ensures the quality of the general outputs.

The third stage is a rather simple one. By downscaling and upscaling the output by 8, the effect of the pixel art style will be more obvious.

Evaluation

Advantages:

- Guided by LoRAs, the model is able to generate various expressions in a more controlled way. Different expressions are also more recognisable and easier to distinguish, as shown in Figure 6.6.
- Generating the image and injecting the prompt two times create a second “insurance” for the overall quality of the output, making it more likely for the player to get the desired images in only one click.
- The output is more consistent. The faces look more belong to one person, reducing the randomness of the basic workflow.

Disadvantages:

- Since the LoRAs are all modifying the facial expression, they influence each other when all loaded. To address this the weights are changed dynamically when making the API call, however, this change will cause the model to reload every time a new expression generation is requested and it slows the general speed. The workflow typically takes around 10-15 seconds to generate a single expression.

- The pixel art style is not as good compared to the vanilla version.

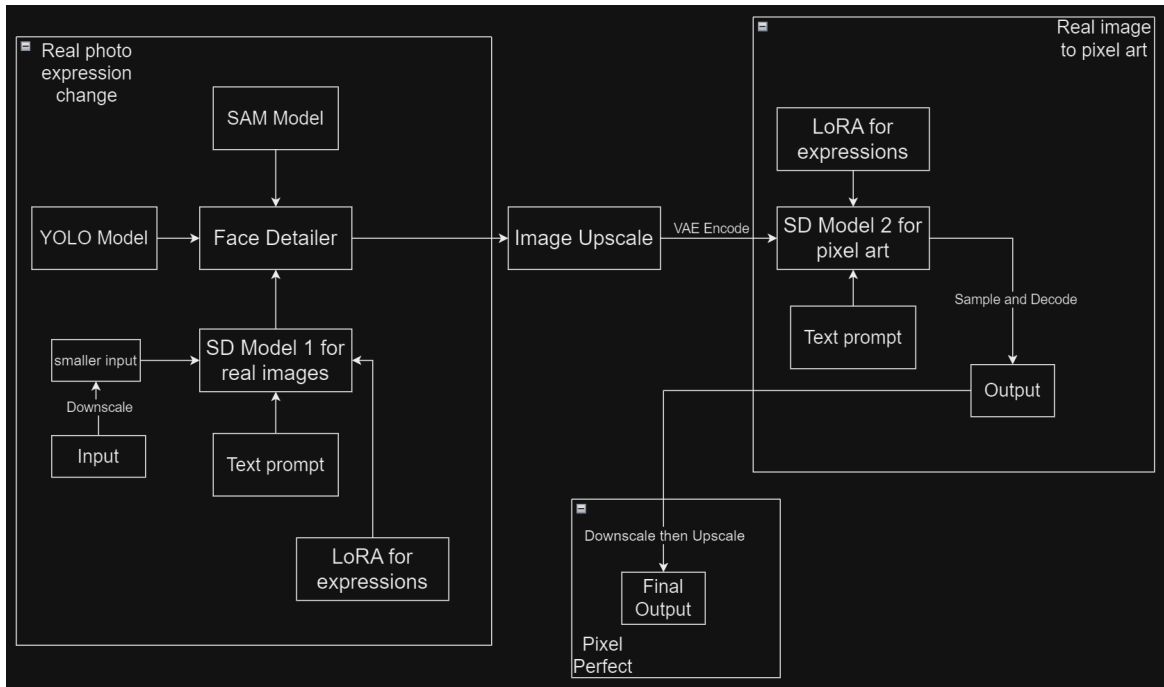


Figure 6.5: Simplified final expression change workflow



Figure 6.6: Different expressions generated by the final workflow

6.1.3 Other Workflows

Some other workflows are also designed, tested and compared. However, due to the limitation of the length and time and the fact that this project is not mainly about expression changing, the detailed introductions and comparisons of these workflows will be skipped. The graphs of two example workflows with great potential are shown. Both of them use ControlNet for image regulation. The workflow shown in Figure 6.7 combines ControlNet with the Face Detailer, while the one in Figure 6.8 uses OpenPose ControlNet, and feeds the encoded image info together with the text prompts to the KSampler to get more control over the facial expression.

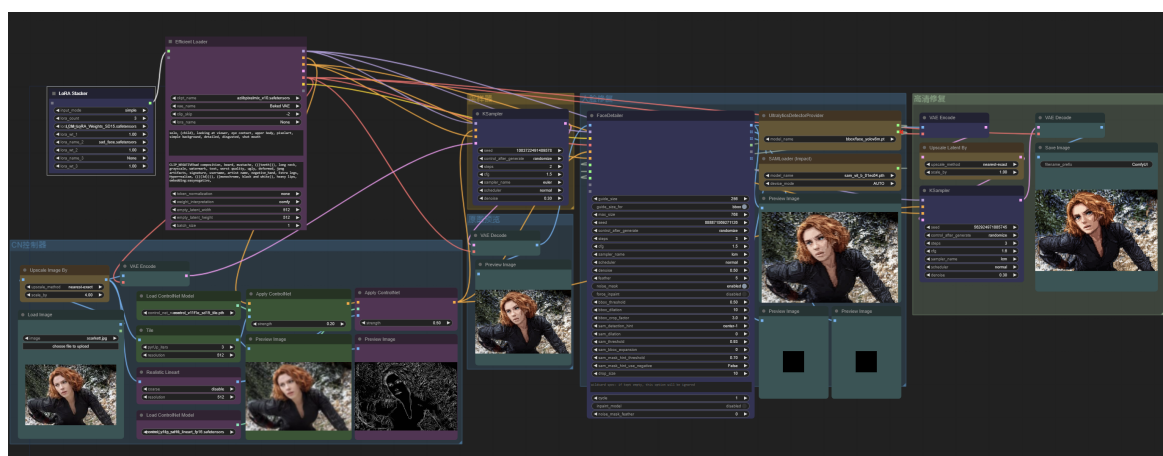


Figure 6.7: Workflow combining ControlNet and Face Detailer

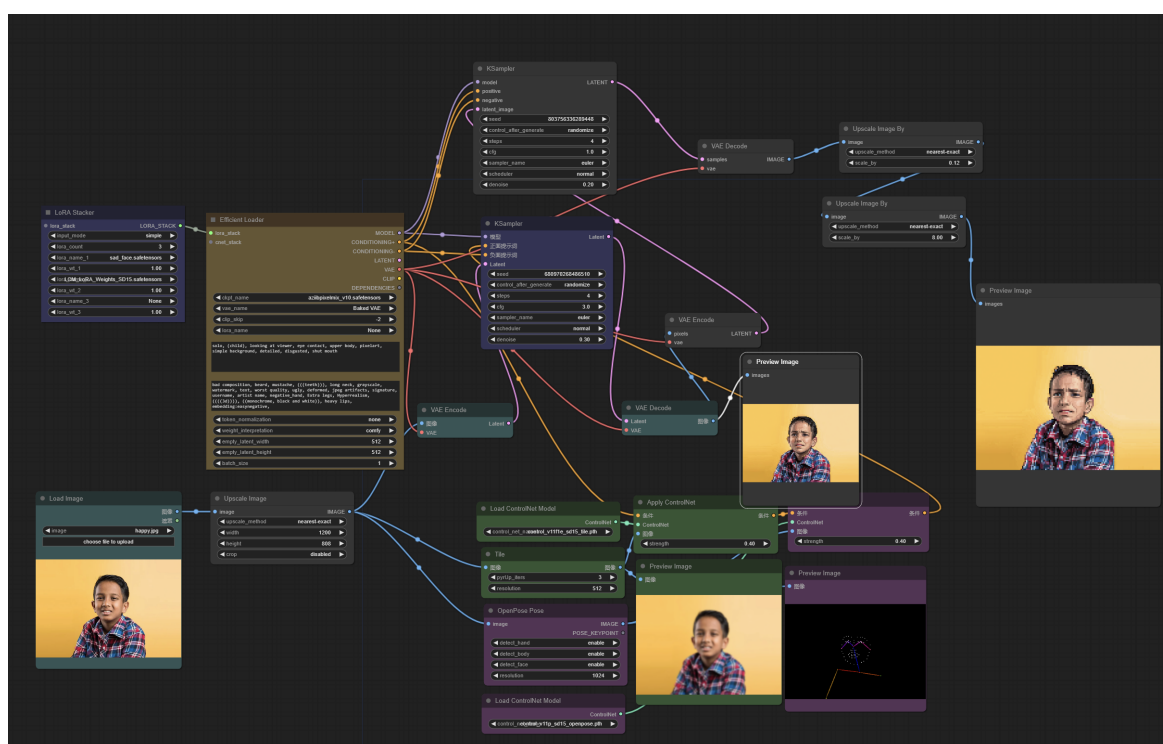


Figure 6.8: Workflow using OpenPose ControlNet and encode that as prompt

6.2 Front-end design

For the players' ease of use, the detailed workflows and parameters are all hidden from them, and only buttons for uploading the image and generating buttons will be given. Figure 6.9 gives a full picture of the photo upload and image generation scene. The functions of the buttons are crystal clear with their names on them. The generate buttons are used to generate the photo with the specified expression, and the show button is used to display the generated button in the display panel, which is the second white box on the right. The generate all button at the bottom

provides a convenient one-click generation, as all the photos are needed for the SAT exercises later in the game. The interface also uses colour coding for different types of expressions: red is for negative emotions, yellow is for neutral ones and blue is for happy. When the photo is not uploaded, only the upload button, back button and continue button are enabled, meaning the user can either upload their photo or choose to leave the scene and continue the game without the images.

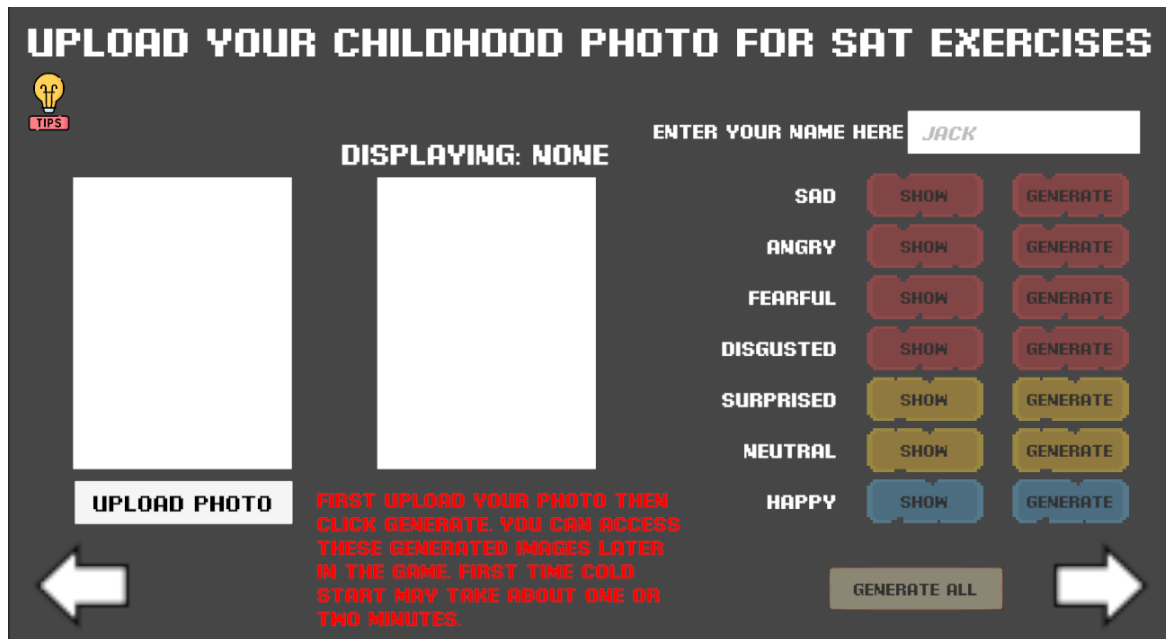


Figure 6.9: Front-end for photo uploading and expression changer

6.3 Cloud Server Setting

With the workflow and front-end ready, players are now expected to call the workflow during the image generation phase. ComfyUI, being an application with pre-defined API calls, can effectively function as a backend server. By hosting ComfyUI on a cloud server, this project facilitates easy API calls and reduces the computational load on the client side. However, the cost of maintaining a GPU server can be quite high, making cost control a significant consideration for this project.

To address this, the project utilizes serverless computing services, specifically the Function as a Service (FaaS) offered by Alibaba Cloud [51]. As discussed in the Literature Review chapter, the pay-as-you-go model of FaaS is particularly cost-effective for deploying large models that require GPU resources, such as Stable Diffusion and ComfyUI. The serverless approach works as follows: when a request is sent to the server, a new instance of the function is created, and resources are allocated automatically (“cold start”). Once the task is completed, if no further requests are received, the instance automatically terminates, thereby freeing up resources and minimizing costs. This model ensures that the project remains cost-efficient while

leveraging powerful cloud computing resources.

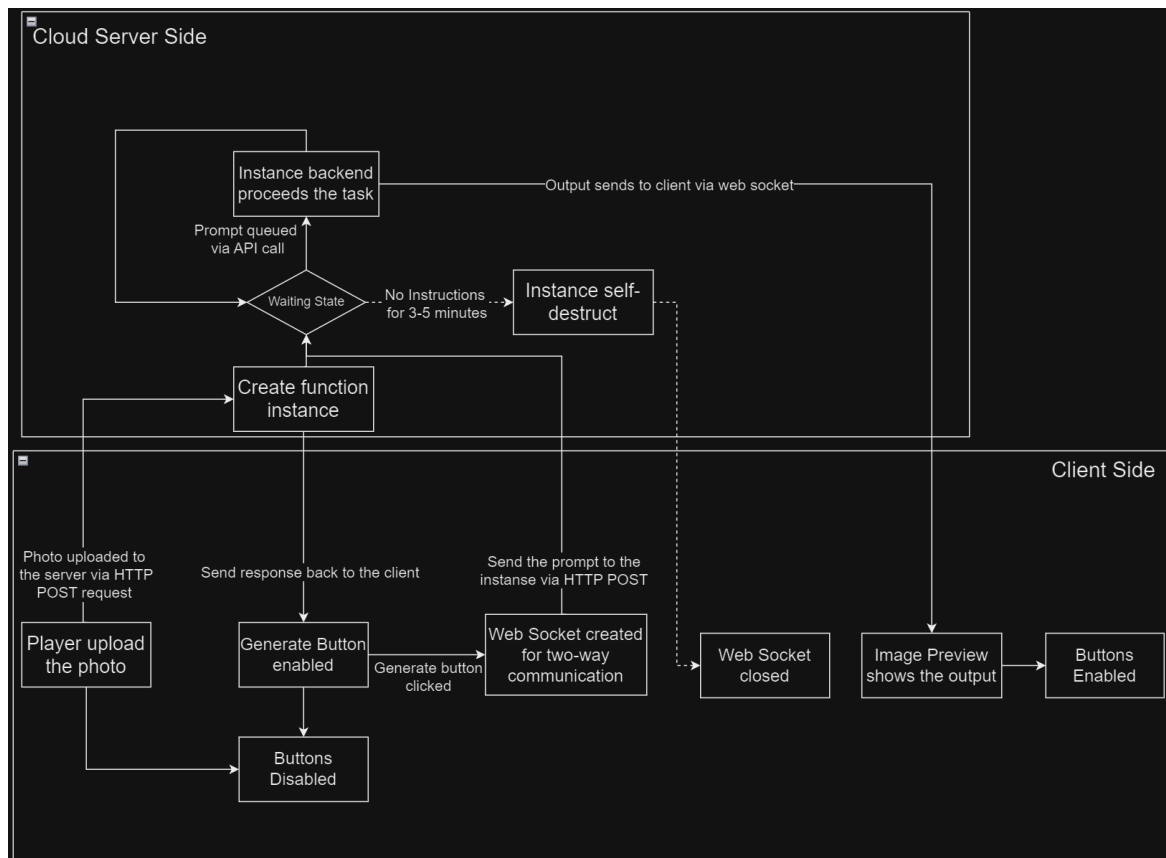


Figure 6.10: Serverless Cloud Server workflow

Figure 6.10 shows the general workflow of how the server is integrated in this project, typically with the front-end. The workflow is split into two halves, with one showing the client side and the other with the cloud server side to provide a clearer view. As said, the player can choose not to upload the photo and skip this process completely. In this case, the cloud server will not be invoked at all, costing nothing at all.

When the players choose to upload their photo, an HTTP POST request will be sent to the server, uploading their image to the server for the workflow to access later. Since this process usually requires the server to “cold start” the instance, it will take a longer time for the first photo upload, typically 40 seconds in this project’s case. To make sure this period is not interrupted and the photo is uploaded to the server before any generation process can take place, all buttons will be disabled. Only when the server sends back the response of the image uploaded the buttons will be enabled again. Meanwhile, the created function instance will be in its waiting state. It will self-destruct when receiving no instructions for a period of time, this is generally 3 to 5 minutes.

If the player clicks on the generate button when the instance is still alive, a web socket will be created to connect the client and the server on the client side. In this way, the client can get the generated image with ease. The generation prompt will then be sent to the server via another HTTP POST request. Upon receiving the prompt, the ComfyUI will get to work in the backend, just as running it locally. The response will be sent to the client side during the whole process of generating, and through checking the response JSON code, the client can know when the generation is finished and retrieve the generated image from a special node from the designed workflow. The generated image will then be shown on the display panel for the player to review. The function instance, on the other hand, will go back to the waiting state and wait for the next prompt or self-destruction.

The generating process typically takes a longer time for the first image generation, as the first-time generation requires the ComfyUI and models to be loaded, which will generally take another 40 seconds. However, the later generations will take only 10 to 15 seconds to proceed, so the overall generation performance is still considerable. Similarly, to avoid interruptions to the process, all the buttons are disabled during the generation process as well, and will be re-enabled when the current task is finished.

6.3.1 Evaluation

One problem currently with the facial expression changer, which can be observed in the previous section, is that when the photo is uploaded but the page is left open for too long without generation, the instance will self-destruct, causing the later clicking on the generation button not to work. In that case, the photo should be uploaded again so another instance will be created for the web socket to communicate with. It can be designed so that the server is kept active by the web socket immediately after the photo is uploaded, in this case, the server is unlikely to self-destruct as long as the player is still in the scene. However, this is a carefully designed result, if the player leaves the scene open for too long, it will burn the resources while not using them, causing meaningless costs.

To sum up, the cloud server runs on a controllable and flexible budget with a reasonable performance. Although having the problem of long “cold start” time and model loading time, adding to a total of around 90 seconds, the overall performance of the facial expression changer on all 7 emotions is considerable, with a total of around 180 seconds, which means 25 seconds for each. Considering its cloud nature and can be run on almost every environment, even with a very out-of-date GPU, this approach is very promising in providing therapies to people who cannot afford an expensive GPU. This exploration makes a great contribution towards making SAT a more accessible psychotherapy tool by providing a solid avatar generation tool with lower requirements.

Chapter 7

Game Design & Software Engineering

7.1 Game Design

7.1.1 General Style Choice

The game is designed to be a 2D top-down one in pixel art style, which highly resembles the famous game *Stardew Valley* [52]. This choice is influenced by the author's personal preference for pixel art games and a strong emphasis on immersion, a central focus of the game design.

Pixel art, with its stylised and simplified visual approach, allows for a unique blend of nostalgia and creative expression. The decision to adopt a 2D style over 3D is based on several key considerations:

1. **Personal Aesthetic Preference:** The author of the project has a deep affection for pixel games, which often evoke a sense of nostalgia and charm. It is also hoped that this artistic style can resonate with many players, creating a warm and inviting atmosphere that can enhance emotional engagement and attachment to the game world.
2. **Immersion Through Simplicity:** The use of pixel art can contribute to a more immersive experience by encouraging players to use their imagination to fill in visual details. Unlike high-resolution 3D graphics, which aim for photorealism, pixel art relies on stylised representations that can evoke powerful imagery without the need for detailed graphics. This abstraction allows players to project their interpretations and emotions onto the characters and settings, potentially deepening their connection to the game.
3. **Avoiding the “Uncanny Valley”:** One of the risks of 3D games is the potential for the graphics to appear “cheap” or unnatural if not executed with high quality. Issues such as rigid animations, awkward character expressions, or inconsistent textures can break immersion and detract from the player's experience. In contrast, 2D-pixel art inherently avoids these pitfalls due to its stylised nature.

4. **Technical Considerations:** Developing a 2D game, particularly in a pixel art style, can be less resource-intensive and more accessible for smaller development teams or individual developers. It reduces the complexity of the animation and rendering processes, allowing the developer to focus more on the narrative and gameplay elements. This approach ensures that the technical aspects, such as character expression control and animation, are more manageable, leading to a more polished and cohesive game experience.

In conclusion, the decision to use a 2D top-down pixel art style for the game is well aligned with the project's emphasis on narrative-driven gameplay and the emotional depth necessary for its themes. By leveraging the strengths of this visual style, the game seeks to create a highly immersive experience that enables players to engage deeply with the narrative. This approach allows players to vividly experience the "virtual trauma" as if it were real, facilitating a meaningful exploration of profound themes such as empathy, healing, and breaking the cycle of violence.

7.1.2 Character Creator

As a common feature in games, the game provides a Character Creator that allows the player to control the look of the protagonist. This feature includes options to control various aspects of the character's outfit, such as the body, hair, torso, and pants, each with one or more colours to choose from. By enabling players to personalize the protagonist, the game aims to enhance player engagement and foster a deeper connection to the character.



Figure 7.1: Character Creator

As shown in Figure 7.1, in the current design, the customization options are straightforward: Players can select different colours for available parts of the character's

outfit, enhancing visual personalization. If a particular body part has only one option, such as the body and the colour of the torsos, the buttons for changing those are disabled (greyed out).

Currently, the game only supports a single gender (male) and one skin colour for the protagonist. Expanding these options will be considered for future development, as adding new character assets and animations requires significant effort and resources. While these limitations might slightly reduce the level of immersion for some players, it is important to note that the protagonist in this game is not necessarily intended to represent the players themselves. This design decision is supported by precedents in other narrative-driven games, such as *Detroit: Become Human* [43], which successfully achieved high levels of player immersion without extensive character customization options.

Overall, the present version of the Character Creator serves as a functional prototype, aligning with the game's focus on narrative-driven gameplay and emotional engagement. It supports the primary goal of exploring how interactive storytelling can simulate “virtual trauma” experiences and provide an effective context for applying the SAT.

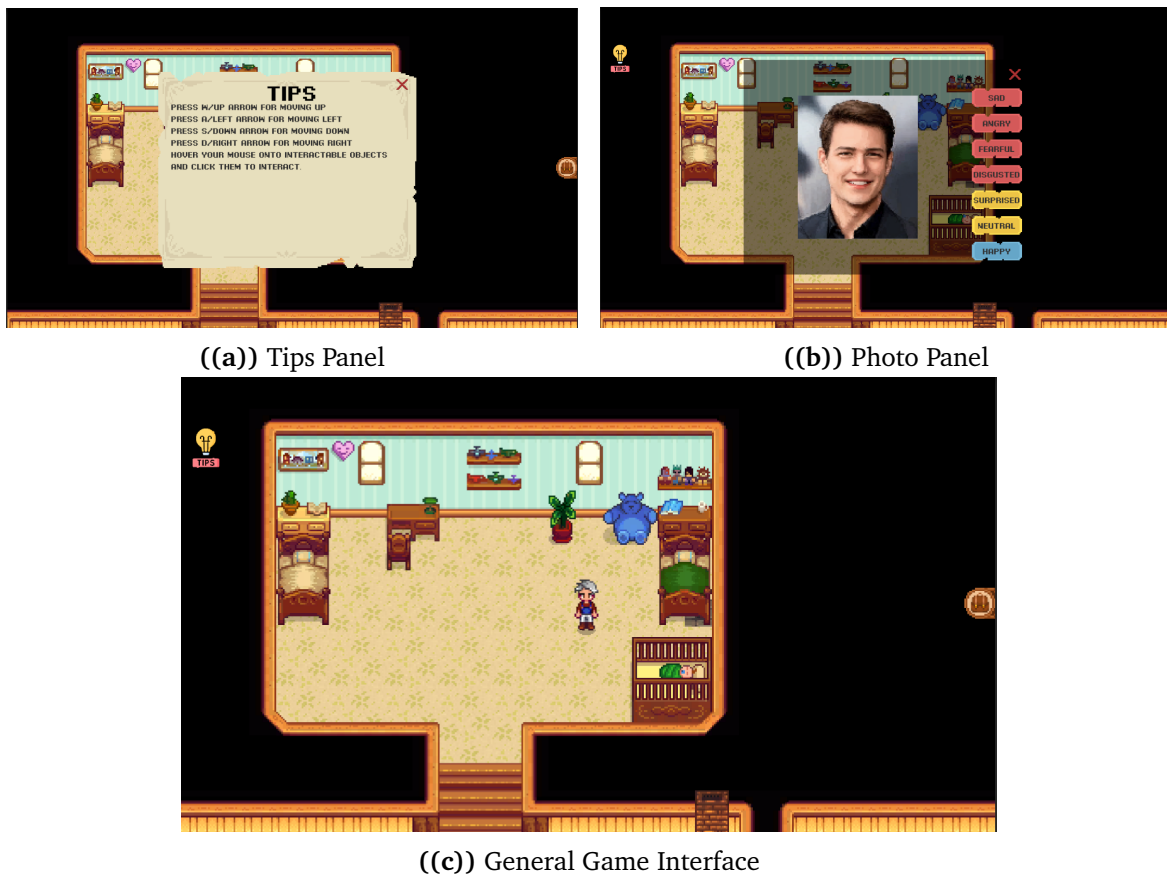


Figure 7.2: Interfaces

7.1.3 Tips Button/Panel

To enhance the player experience and provide clearer guidance, a Tips Button has been added to certain scenes in the game, positioned in the top-left corner, as shown in Figure 7.2(c). This feature was implemented in response to feedback from user evaluations, where many players reported difficulties in progressing through specific scenes.

The game's story design intentionally incorporates a more subtle form of instruction, offering “soft-guidance” to maintain realism and encourage exploration. For example, crucial information might be hidden within interactions with NPCs or embedded in environmental narratives. Although this approach aims to create a more immersive and realistic world, it has the potential downside of causing frustration for players who may not be as engaged in exploratory gameplay or who prefer more direct guidance.

To address this issue and ensure a smoother gaming experience, the tips button and panel (see Figure 7.2(a)) were introduced. By providing context-specific hints, this feature helps players who might otherwise become stuck, preventing potential disengagement from the game. Although this comes with a slight trade-off in terms of immersion—since explicit tips can break the illusion of a seamless game world—it is a necessary compromise to maintain overall player enjoyment and ensure the game remains accessible to a wider audience.

7.1.4 Photo Button/Panel

By clicking the Bag icon on the right of the screen, the panel for displaying the previously generated image will be opened. As shown in Figure 7.2(b), the player can choose to view different emotions just as in the Expression Changer scene.

7.1.5 Cursor Implication

As a narrative-driven game that emphasizes interactive storytelling, player interaction with the game world primarily occurs through mouse clicks. To enhance the user experience, a Cursor Manager has been implemented as part of the game's UI system. The Cursor Manager serves several key functions to make the UI more ac-

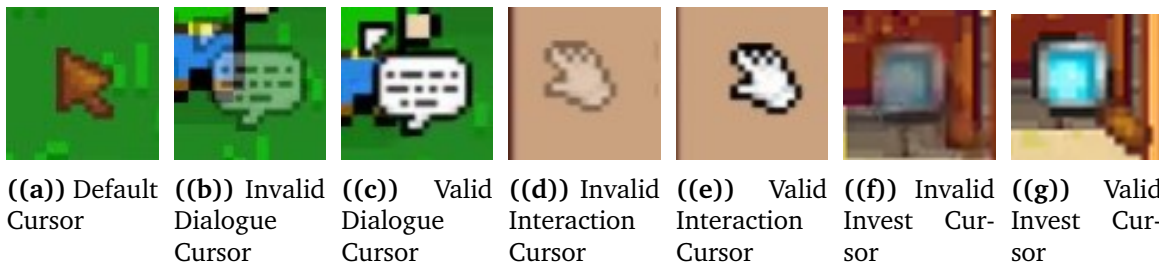


Figure 7.3: Cursors used in this game

cessible and intuitive:

1. **Visual Feedback for Interactions:** The cursor manager dynamically updates the cursor texture based on the context of the potential interaction. When hovering over different interactable objects, the cursor changes to indicate the specific type of interaction available, such as investigating an object, opening a door, or initiating dialogue with a character, as shown in Figure 7.3(c), 7.3(e) and 7.3(g). This visual feedback helps players quickly understand their options and encourages exploration within the game world.
2. **Distance-Based Interaction Indicators:** In addition to showing the type of interaction, the cursor manager also indicates whether the player is within the necessary range to perform the action. If the player is too far away from an interactable object, the cursor will turn somewhat transparent to signal that the action cannot be performed from the current distance. Figure 7.3(b), 7.3(d) and 7.3(f) gives an illustration of this feature. This feature prevents confusion and unnecessary frustration by clearly communicating the requirements for interaction, encouraging players to move closer to objects or characters they wish to interact with.

By providing clear visual cues through the cursor, the system lowers the cognitive load on players, making it easier for them to navigate the game world and understand potential actions. This is particularly important in a narrative-driven game where maintaining immersion and smooth gameplay flow is crucial. The cursor manager ensures that players can easily engage with the story without being hindered by unclear or complex controls.

7.2 Software Engineering

The development of the game involves experimenting with different software engineering techniques. In the following, the section will cover the comparison of two different ways of scene loading mechanisms used in the development and justify why additive loading is preferred. The section will also cover the evolution from using direct method calls between different components in the game to using a centralized event handler, and what benefits it brings. The section will also briefly introduce the animation manager it uses, which is a self-invented tool for playing cutscenes with dialogues. The Animation Manager's special structure makes an "accidental" speed-up for scene changing in certain cases, making the game smaller in size.

7.2.1 Scene Loading Mechanism

In game development, particularly in Unity, scene management is a critical aspect of ensuring smooth transitions, efficient resource use, and player immersion. During the development of this project, both Traditional scene loading and Additive scene loading were tested, and the latter was chosen as the preferred approach due to its flexibility and benefits for the game's design.

Traditional Approach

Traditional scene loading, also referred to as single scene loading, is a straightforward method of loading scenes in Unity. In this approach, each scene is loaded in its entirety while the previous scene is unloaded from memory. This method effectively clears the game's memory of the old scene's assets and ensures that only the assets of the currently active scene are loaded and in use. Traditional loading works best for level-based games and those with clear scene boundaries.

Advantages:

- **Ease of use:** Traditional scene loading is easy to implement and understand, as it completely separates different game areas or levels.
- **Memory Management:** By fully unloading a scene, the game frees up memory and resources, which provides an easy way of memory management.

Limitations:

- **Loading delays:** Traditional scene loading can result in noticeable delays as the game transitions between scenes, which can break immersion, particularly in seamless open-world games or RPGs.
- **Redundant asset loading:** In the traditional method, each scene includes its own copy of common assets (UI, player character, event system, etc.) When switching scenes, the game unloads all assets from the previous scene and loads all assets from the new one, even if many of those assets are identical, which can increase memory usage and cause longer load times.
- **Loss of Persistence:** One major drawback is the lack of persistence between scenes. If the player or other assets need to remain consistent across scenes (e.g., the player's position or game state), additional effort is required to save and restore those elements, which can complicate development.

Additive Scene Loading

In additive scene loading, several scenes can be loaded together, and they will all be active in memory. The different scenes will serve different purposes, allowing for a seamless transition between different areas of the game without having to fully reload. By having a “persistent scene” in the game that is loaded all the time in the game, this approach makes it possible to have the special assets, such as the player character or global managers, to persist across multiple scenes without the need for reloading.

Advantages:

- **Persistence Across Scenes:** Additive scene loading allows for the creation of persistent scenes, such as player scene, manager scene and UI scene, that remain loaded while only adding or removing the environment scenes. In this project, there is a persistent scene that manages the player and other global

managers, together with a UI scene for the buttons, panels discussed in the previous sections and a loading image. This ensures that these components remain consistent across different environments or locations in the game.

- **Flexibility:** With additive loading, different components of the game (e.g., environments, NPCs, UI elements) can be managed in separate scenes and loaded as needed, which can simplify the management of complex game systems and improve performance.

Limitations:

- Managing multiple scenes and ensuring proper communication between them can increase the complexity of the game's architecture, compared to the naive single scene loading approach.

For this project, with a 2D top-down RPG-like gameplay in the first stage, additive scene loading was chosen due to its ability to manage a persistent scene and a UI scene for the player and other global assets. The persistent scenes remain loaded throughout the game, ensuring that the player's position and other assets are maintained as the player moves between different scenes, avoiding redundant asset loading at the same time. The RPG nature of the game in the first part, where players frequently transition between different environments, makes additive loading particularly effective in reducing the need for saving and reloading the player's state.

7.2.2 Centralized Event Handling System

Direct Method Calls

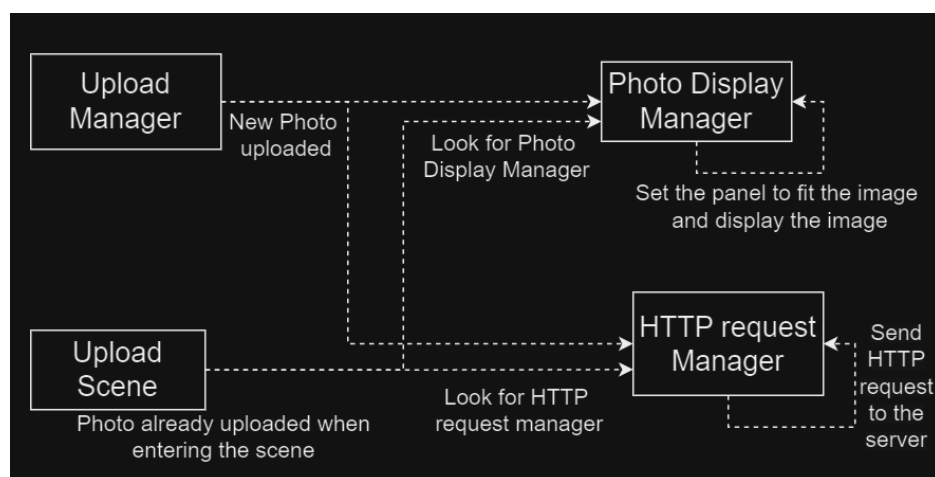


Figure 7.4: Example direct method call workflow for photo upload

During the initial stages of development, the game architecture relied on direct method calls between components, such as global managers, to handle interactions and game events. Figure 7.4 illustrates a typical interaction in the process of uploading an image. In this example, two situations trigger the HTTP request and

subsequent actions: either a new photo is uploaded, or the photo has already been uploaded when entering the photo upload scene. In both cases, the Photo Display Manager and HTTP Request Manager are fetched directly from the scene, and their respective methods are executed directly.

While this approach is simple and effective in small-scale projects, it quickly became apparent that such direct method-calling between components led to tightly coupled systems. Each manager or component depended directly on the others, making the game's architecture rigid and difficult to modify or extend as the project grew more complex. As the project grew, managing these direct connections became increasingly challenging, making it harder to debug and maintain the code efficiently. What is more, directly fetching components from the scene may not always be reliable. If certain managers were not present in a particular scene, this would result in an "Object Not Found" error, requiring additional safeguards in the form of conditional checks. This added unnecessary complexity and inefficiency to the code.

Centralized Event Handling

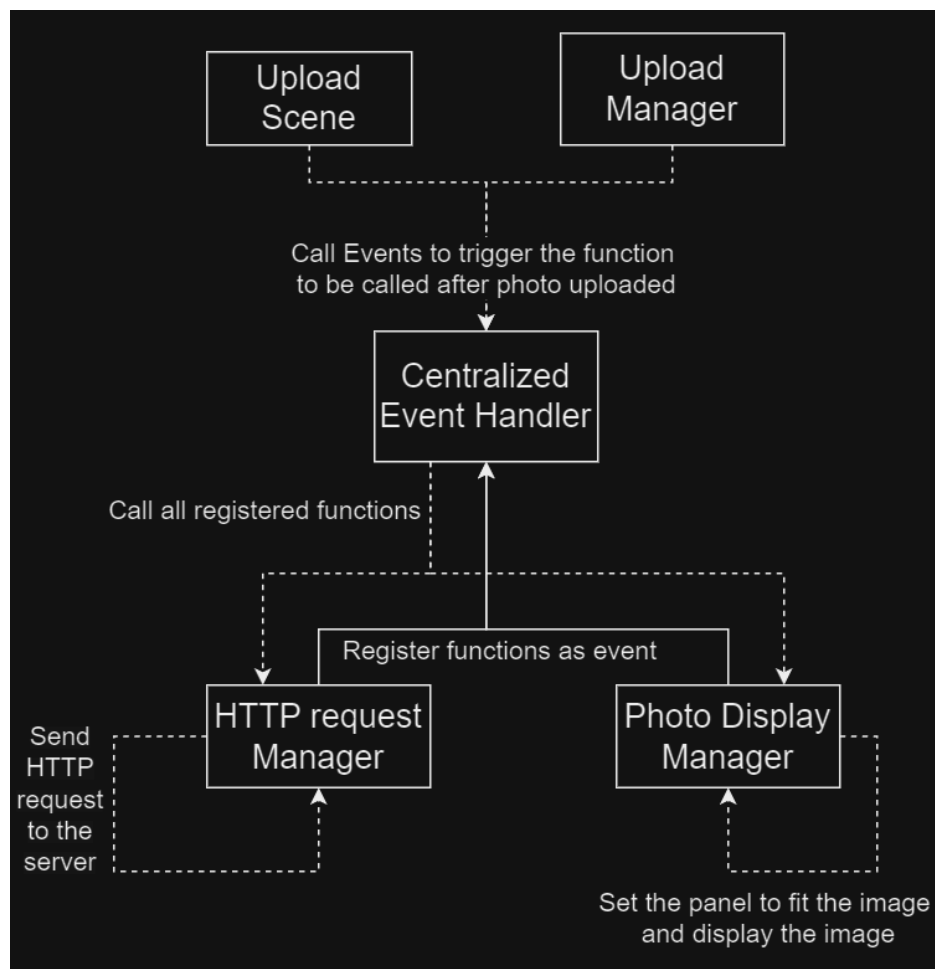


Figure 7.5: Example centralized event handler workflow for photo upload

To resolve these issues, the project was restructured to implement a Centralized Event Handling System. In this new architecture, components no longer directly interact with each other. Instead, they communicate indirectly through an event-driven model, which decouples the components and allows for more flexibility.

Figure 7.5 shows the example photo upload workflow under such a structure. Each component can register to events in the event handler. In this example, the HTTP Manager registers the function of sending a request to the server to the OnPhotoUpload Event, and the PhotoDisplayManager does this for setting the photo panel size and displaying the image. In this case, when a new photo is uploaded or the scene is entered with a photo already uploaded, it will call the event in the event handler, instead of calling the functions from other components directly. The event will then be broadcast from the event handler, and any interested component (e.g., the Photo Display Manager and HTTP Request Manager in this case) can listen for and respond to these events. This shift has introduced several benefits:

- **Loose Coupling:** The centralized event handler looses the coupling between the components and eliminates the need for systems to know about each other's existence. For instance, the Upload Manager no longer directly calls methods in the HTTP Request Manager. Instead, the latter subscribes to relevant events and reacts accordingly, which allows each component to evolve independently.
- **Modular Functionalities:** With this event-driven approach, each function has become more modular, which makes adding or removing features become more manageable. New functionalities can be integrated by simply subscribing to the necessary events without modifying the existing components, significantly reducing the risk of introducing bugs. This also mitigates the issue of fetching non-existent components from scenes. Since events are broadcast globally, components only need to subscribe to the events they are interested in, eliminating the need for scene-specific component fetching and reducing the risk of "Object Not Found" errors.
- **Scalability and Flexibility:** The event-driven architecture is inherently more scalable. As new features or systems are added to the game, they can simply hook into the existing event system. This also makes it easier to extend or modify game mechanics without modifying large portions of the code.

The transition from a tightly coupled, direct method-calling approach to a centralized event-driven architecture has greatly improved the flexibility, maintainability, and scalability of the game. By using an event handler to manage interactions and communication between components, the game architecture is now better suited to accommodate future expansions and changes while reducing the risk of errors and increasing development efficiency.

7.2.3 Animation Sequence and Manager

Unity typically utilizes the integrated Timeline system for playing cutscenes in the game. By playing different animations in parallel or in sequence, it can allow the de-

velopers to create cutscenes with ease. However, the vanilla version of cutscenes provided by Unity itself does not support dialogues, which is not expected in a narrative-driven game, where the story is mainly developed through dialogues and other forms of narrative. To make auto-playing cutscenes with dialogues, the animation sequence and animation manager are developed. Through the help of these tools, the game is able to create easy cutscenes allowing characters to move, have dialogues, display animations and modify their status. In addition, the Animation Manager's structure, combined with the repetition of the environment scene, led to a more efficient way of changing the scene when doing SAT exercises.

Animation Step

An animation step is the minimal unit in this framework, capturing what type of animation the specific NPC should play. It has 7 different types, each having a different logic for the manager to process. The 7 types typically involve:

1. **Movement:** By setting the waypoints in the animation step, it will allow the specified NPC to move along these waypoints.
2. **Emote:** By setting a pre-defined animation trigger's name, it will make the specified NPC play the given emotion animation clip.
3. **Wake:** This will cause the Player to wake, which is specifically used in the starting scene.
4. **Conversation:** This will call the implemented dialogue system and display the pre-set dialogues when executed.
5. **SceneChange:** This will unload the current environment scene and load the target scene specified in the Animation Step's field.
6. **CutSceneTrigger:** This will play the next Unity Timeline animation in the scene. This is specifically used for the several years later scene where images and dialogues are required to be displayed in sequence.
7. **None:** This is an animation type that only deals with the inner state of the specified NPC, such as some boolean parameters.

Animation Sequence

The Animation Sequence contains a list of Animation Steps, and this is what the "animation" or cutscene actually is in this game. The Animation Sequence is made a Scriptable Object in Unity and each animation is an object containing sequences of Animation Steps. By feeding the Animation Steps in its list one by one, the mechanism successfully achieves the goal of having a continuous animation which includes interactable dialogues.

Each sequence has a unique ID. When an Animation sequence is played it will be recorded and usually, the played sequence will not be played again. This is very useful for situations, where for example, the players enter a scene with some starting cutscene and re-enter the scene again. As the same cutscene is normally not expected to be played twice.

Animation Manager

Animation Manager is the class for eventually executing the Animation Steps, which are fed in by the Animation Sequence. This Animation Manager object is added to every scene that requires cutscenes to be played. With a starting sequence (or can be set to “None” if not wanted), it will play a sequence when the scene is first loaded. Another way of triggering a cutscene is when the player enters specific areas.

Special Scene Load Mechanism

In the SAT exercises stage, most of the exercises take place in the same environment, typically the player’s house. For many of the exercises, the environment will remain completely the same, changing only the starting sequence given in the Animation Manager in that scene. As a result, a new way of swapping the scenes is designed: Instead of changing the entire scene, with potential redundancy for the entire environment, causing extra storage taken by those scenes, some template scenes will be created. By inserting different starting sequences into its Animation Manager, it will be more efficient in loading and storage-wise, while functioning the same.

Chapter 8

User Evaluation

Given the dual purpose of this project – serving as both a therapeutic and educational tool – evaluating its effectiveness requires considering multiple aspects. At the core, the game’s usability is the foundational element that ensures players can comfortably engage with the gameplay and interface without any confusion or barriers. Usability is the most basic and essential requirement for the success of the game; without an intuitive and user-friendly interface, players would face challenges engaging with the more advanced features, including the Self-Attachment Technique (SAT) exercises.

The psychological benefits of the game are also evaluated. This involves assessing how effectively the game helps players engage with Self-Attachment Technique (SAT) exercises and how well it fosters empathy and emotional healing. Given the difficulty of finding individuals who have experienced real genocide trauma for the trials, this phase did not extensively assess the therapeutic impact of the game on trauma survivors. However, it is worth noting that the influence of SAT as a therapeutic tool has been examined in a broader sense, so the therapeutic effect should be promising.

As a result, the primary focus of this user evaluation was on studying the game’s usability and educational effects. To achieve this, participants were asked to fill out questionnaires both before and after playing the game, and 21 responses were obtained when this part was being written. This allowed for a comprehensive study of how each component performed, as well as an assessment of the overall effectiveness of the project in conveying its educational aims.

8.1 Evaluation Questionnaire

The questionnaire can be found on https://imperial.eu.qualtrics.com/jfe/form/SV_0JycpkTfZtmkTMa.

As discussed, the questionnaire is designed to cover participants’ evaluation of the game from two aspects: Usability and Effect of Education. To better evaluate the

latter, the questionnaire is divided into two parts: one to be done before playing the game and the other afterwards.

8.1.1 Part I: Pre-Game Evaluation

The first part focuses on participants' understanding of key concepts used in this project, such as the cycle of violence and the Self-Attachment Technique (SAT). To ensure clarity, a brief introduction to these terms was provided along with the questionnaire to prepare participants for the evaluation. Table 8.1 shows the responses received from the 21 participants. As expected, very few participants had experienced severe childhood trauma, and while most had heard of the term cycle of violence, only a small number were familiar with SAT. Fortunately, most participants expressed a positive response to the provided explanations, indicating a better understanding of these concepts after reading the descriptions.

Table 8.1: Responses received from participants for Part I

Statement/question	Yes	No	Not Sure
Have you experienced any trauma or significant emotional pain during your childhood that still affects you today?	2/21	15/21	4/21
Have you heard of the term "Cycle of Violence" before?	9/21	11/21	1/21
Do you feel that you have a better understanding of the term "Cycle of Violence" after reading the introduction?	17/21	4/21	0/21
Do you believe that breaking the cycle of violence requires completely giving up on seeking revenge, even against those who have caused you significant harm?	1/21	11/21	9/21
Have you heard of the Self-Attachment Technique (SAT) before?	2/21	19/21	N/A
Do you feel that you have a better understanding of SAT after reading the introduction?	13/21	1/21	7/21

Table 8.1: Responses received from participants for Part I (continued)

Statement/question	Yes	No	Not Sure
Do you enjoy playing games?	11/21	2/21	8/21 (Occasionally)
Do you like visual novels? (or narrative-focused games)	12/21	4/21	5/21

It is important to note that few people agree that breaking the cycle of violence requires completely giving up on seeking revenge, particularly against those who have caused significant harm. This insight supports the case study from Chapter 3 and reinforces the story design decision to allow Ian to seek revenge against Morris. By incorporating this element, the narrative avoids the common pitfall of focusing solely on the forgiveness vs. revenge debate, and instead highlights the theme of breaking the cycle of violence. This approach was well-received by participants, as it acknowledges the complexities of human emotions in response to trauma.

The results of Part I show that most participants were adequately prepared for the game trial and provide evidence to support the narrative choices made in the game.

8.1.2 Part II: Post-Game Evaluation

The second part of the questionnaire evaluated both the usability of the game and participants' overall feedback on the gaming experience, particularly the storytelling and engagement levels.

Table 8.2 summarizes the participants' responses, showing that most participants found the software easy to use. Furthermore, 13 participants reported finding the game engaging, while only one participant felt the game was not appealing. The Facial Expression Changer was also well-received, with participants giving it an average performance rating of 7.67 out of 10 for image quality. While the waiting time for image generation was not rated as highly, it remained within a reasonable range of satisfaction.

The effect of the game is also promising, with 13 of the participants saying that the game has reminded them of the fact that focusing on revenge only will cause the cycle of violence, and it is through peace and healing that can we break free. It is also worth noting that among the 21 participants, 6 of them did not complete the game due to various reasons. The result shows this has a great impact on their choices towards the question in Figure 8.1, as most of them chose the 3 choices from the bottom ("not sure" and revenge should be considered).

One typical feedback received from the participants is that the game is too text-intensive and the game took too long (typically one hour) to finish. According to the survey, only 9 participants read the full story while others all skipped some parts of

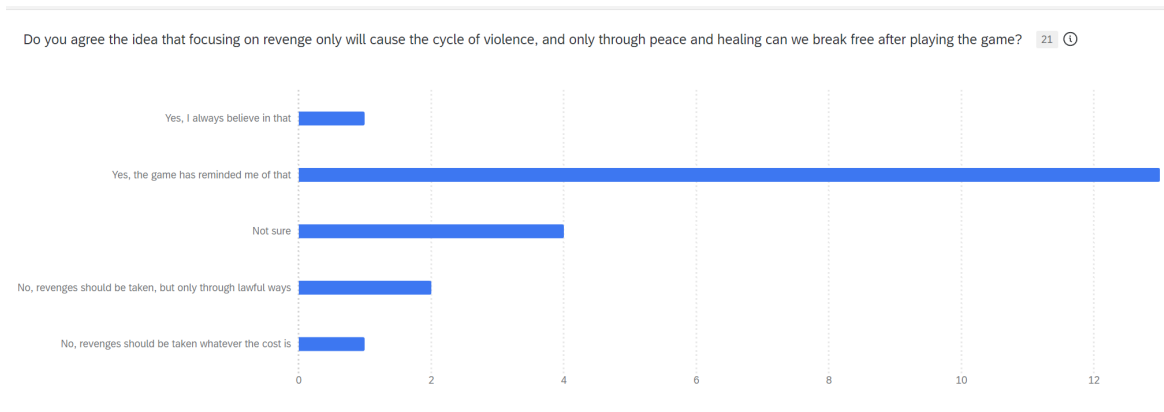


Figure 8.1: Result on the effect

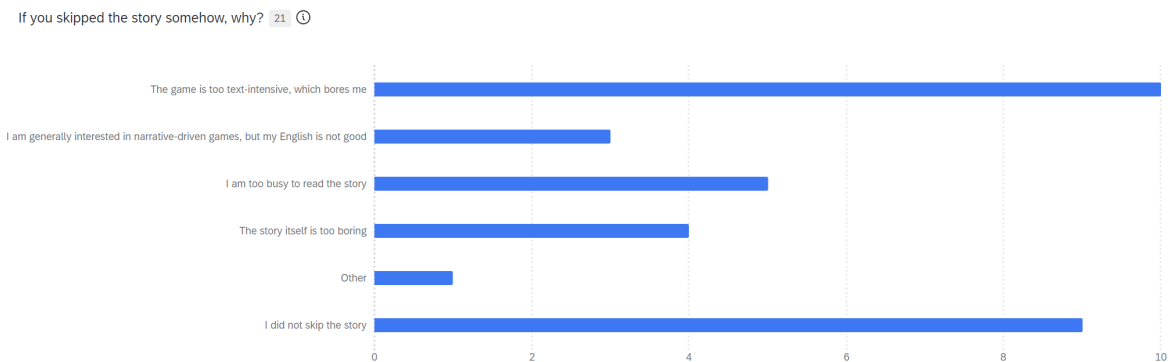


Figure 8.2: Result on reasons for skipping stories

the story to some extent. As shown in Figure 8.2, the most common reason is the text is too long in the game. Other reasons may include the game being too long for participants or the story being too boring to them.

Table 8.2: Responses received from participants for Part II

Statement	Strongly Agree	Agree	Neither Agree Nor Disagree	Disagree	Strongly Disagree
The overall experience of the game is satisfying.	1/21	15/21	4/21	1/21	0/21
The game mechanics are easy to understand and I did not have any trouble during the process.	11/21	10/21	0/21	0/21	0/21
I felt engaged when playing the game.	1/21	12/21	7/21	1/21	0/21

Table 8.2: Responses received from participants for Part II (continued)

Statement	Strongly Agree	Agree	Neither Agree Nor Disagree	Disagree	Strongly Disagree
The performance of the Facial Expression Changer is good (waiting time is not too long).	0/21	12/21	7/21	2/21	0/21
The output of the Facial Expression Changer is satisfying.	2/21	18/21	1/21	0/21	0/21
The protagonist in the game is a total coward.	0/21	4/21	9/21	6/21	2/21

8.2 Conclusion

The user evaluation shows promising potential for this project. The overall experience was generally positive, with participants finding the game intuitive and easy to use. The narrative was effective in creating an immersive world, allowing players to explore the theme of breaking the cycle of violence. The Facial Expression Changer was widely praised as an innovative and enjoyable component of the game, with its performance recognized for both image quality and time-efficiency.

However, some drawbacks were identified, particularly concerning the game's text-heavy format, which may reduce player engagement. The quality of the storytelling can also be improved by providing the players with a more charming story in a more interesting setting. These valuable insights provide clear guidelines for future improvements, ensuring that the game continues to evolve and better meet the needs of its audience.

Chapter 9

Conclusion & Future Work

9.1 Conclusion

This project provides a solid prototype that aims to serve two distinct purposes: as a therapeutic tool for individuals who have experienced genocide trauma, and as an educational tool for a broader audience to foster compassion, empathy, and reflection on conflict resolution. By combining the Self-Attachment Technique (SAT) with narrative-driven gameplay and Stable Diffusion, this project extends the application of SAT beyond its traditional use as a therapeutic tool, showcasing its potential in educational contexts.

Through immersive storytelling and SAT exercises, the game creates a virtual experience that enables players to engage with complex emotional challenges in a controlled environment. This not only helps genocide survivors with emotional healing but also educates players on the psychological effects of violence, encouraging them to consider their role in promoting peace and breaking the cycle of violence.

Through immersive storytelling and SAT exercises, the game creates a virtual experience that enables players to engage with complex emotional challenges in a controlled environment. This not only helps trauma survivors with emotional healing but also educates players on the psychological effects of violence, encouraging them to consider their role in promoting peace and breaking the cycle of violence.

The integration of Stable Diffusion for personalized, dynamic visual feedback further enhances the game's potential as a self-administered therapeutic tool, while the game's accessible design ensures that it can be used by a wide range of players. Although the full therapeutic impact of the game will require further evaluation with participants who have experienced real trauma, the project already demonstrates promising potential in education through interactive gameplay.

9.2 Future Work

While this project has provided a solid prototype for exploring the potential of SAT, especially in the educational field, there are several areas for improvement and further study:

1. **Better storytelling:** As identified in user feedback, the game's text-heavy format and storyline could benefit from further refinement. Future experiments can focus on creating a more engaging and dynamic narrative, with a balance between text and interactive elements, as well as exploring more immersive settings to keep players invested in the story.
2. **Audio and Cutsscenes:** The current version of the game does not yet include audio, which offers an opportunity to further enhance immersion. Adding suitable background music, sound effects, and voiceovers could deepen the player's experience, creating a more engaging atmosphere. Additionally, introducing more animations and cutsscenes during dialogues would add variety and help maintain the player's interest throughout the narrative.
3. **Expand Protagonist Customization:** While the game includes a Character Creator, the current customization options are relatively limited. Players can choose from three hairstyles, three types of torsos, and two types of pants, each with different colour options. Additionally, the protagonist is currently a male character with a fixed skin colour. These limitations are due to the focus on other core development in the project, and leaving the potential to expand customization in future updates. Adding options such as female characters and more diverse skin tones could enhance player connection with the protagonist and offer a more personalized gameplay experience.
4. **Better Facial Expression Changer:** While the Facial Expression Changer received positive feedback during user evaluations, there is still room for enhancement. Exploring more advanced workflows could yield more realistic outputs and more lively expressions. Additionally, as of the completion of this report, the FLUX image generation model has garnered significant attention for its versatility in producing highly realistic images. This emerging technology could serve as a cornerstone for future improvements, providing more sophisticated and authentic facial expressions for use in the game.
5. **Evaluation of Therapeutic Effects:** It is mentioned in the previous chapter that due to difficulties in finding participants with real genocide trauma, the game's effectiveness as a therapeutic tool has not been fully assessed. While the effectiveness of SAT in psychotherapy has been proven in other contexts, it is suggested to conduct further studies to test its specific application within this game, particularly for trauma survivors.

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